Deep Learning - Based Detection of Renal Stones in Ultrasonography Images

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By

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CERTIFICATE

This is to certify that the thesis entitled “Deep Learning - Based Detection of Renal Stones in Ultrasonography Images” submitted by Patel Alauki Manish (Roll No.: 24MD002), in partial fulfillment of the requirements for the award of the degree of Master of Technology in Data Science at Pandit Deendayal Energy University, Gandhinagar, is a bonafide record of work carried out under my supervision and guidance.

To the best of my knowledge, the work presented in this thesis has not been submitted elsewhere for the award of any other degree or diploma.

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DECLARATION

I hereby declare that the thesis entitled "Deep Learning - Based Detection of Renal Stones in Ultrasonography Images" submitted to Pandit Deendayal Energy University, Gandhinagar, in partial fulfillment of the requirements for the award of the degree of Master of Technology in Data Science, is a record of original work carried out by me under the guidance of Dr. Yogesh Kumar.

I further declare that this work has not been submitted elsewhere for the award of any other degree or diploma. All sources of information used in this thesis have been duly acknowledged.

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ABSTRACT

This thesis focuses on the automated detection and localization of renal stones in ultrasonography images. The primary objective of this work is to develop a deep learning-based system for accurate detection and localization of kidney stones, along with improved generalization on real clinical data. The study utilizes publicly available ultrasound data and real clinical data, along with preprocessing techniques such as median filtering and CLAHE, data augmentation, and deep learning approaches.

The proposed methodology includes preprocessing, pseudo-labelling, data augmentation, model training, and evaluation, along with Grad-CAM for explainability. Various models such as CNN2D-ResNet, EfficientNet-B0, YOLOv8 (m and l), and RT-DETR were implemented and evaluated using performance metrics like accuracy, precision, recall, and mean Average Precision (mAP).

The results demonstrate that detection-based models outperform classification methods by providing both accurate detection and localization of renal stones. The proposed approach shows improvement over existing methods in terms of robustness to noisy ultrasound images and enhanced performance on real clinical data.

This work contributes to the development of an end-to-end automated renal stone detection system and provides scope for future enhancements in real-world size estimation, advanced explainability techniques, and clinical validation.

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Chapter-1 Introduction

This chapter provides an overview of a problem, need of the study, its goals, the scope and limitations of the work being proposed. This chapter will help understand why this research is important.

1.1 Background of the Study

Nephrolithiasis or kidney stone disease is one of the most prevalent urological diseases affecting a large number of people throughout the world. It is caused by the development of hard, crystal-like deposits (or stones) of either minerals or salts in the kidneys. The actual size, shape and composition of the stones can vary from each other, and when they form, they often cause significant pain in the abdomen (the area between your chest and your legs), obstruction of the urinary tract, infections, and eventually kidney impairment. There has been a significant increase in the number of people with this disease over the last couple decades. This increase can be attributed to changes in our diet, increased of sedentary behaviour, not drinking enough fluids, and environmental conditions. This rise in the number of people developing kidney stones has made it very important for all health care systems in the world to effectively diagnose and treat the disease [1],[2],[3].

One main problem when it comes to kidney stones is that they happen again frequently to people who have had them before. This causes higher healthcare costs for both the patient and provider because there is also more pain associated with having these stones repeatedly occur in the same patient. Early detection and accurate diagnosis are therefore important for preventing problems related to the condition and allowing for effective planning of treatment options. If you can identify a stone that is in a patient's kidney before it causes pain, then this gives the physician an opportunity to provide an appropriate level of treatment through medication and/or the use of minimally invasive procedures (such as percutaneous-lithotripsy), which will reduce the need for complicated surgical procedures and improve patient outcomes.

Kidney Stones diagnosis relies heavily on medical imaging, which typically includes a variety of imaging modalities such as CT scans), X-ray imaging, and ultrasound imaging. CT scans are often thought of as the gold standard because they have the highest degree of accuracy when diagnosing the presence of kidney stones. Unfortunately, CT scans are very expensive, and they expose patients to ionizing radiation. Therefore, they are not a suitable option for repeated use (especially among vulnerable populations like pregnant women and children.) While X-rays are less costly than CT scans, they also have the limitation that there is only a fair degree of sensitivity for the detection of kidney stones and many X-rays will not be able to detect small kidney stones or those made from lower-density materials. [2],[3].

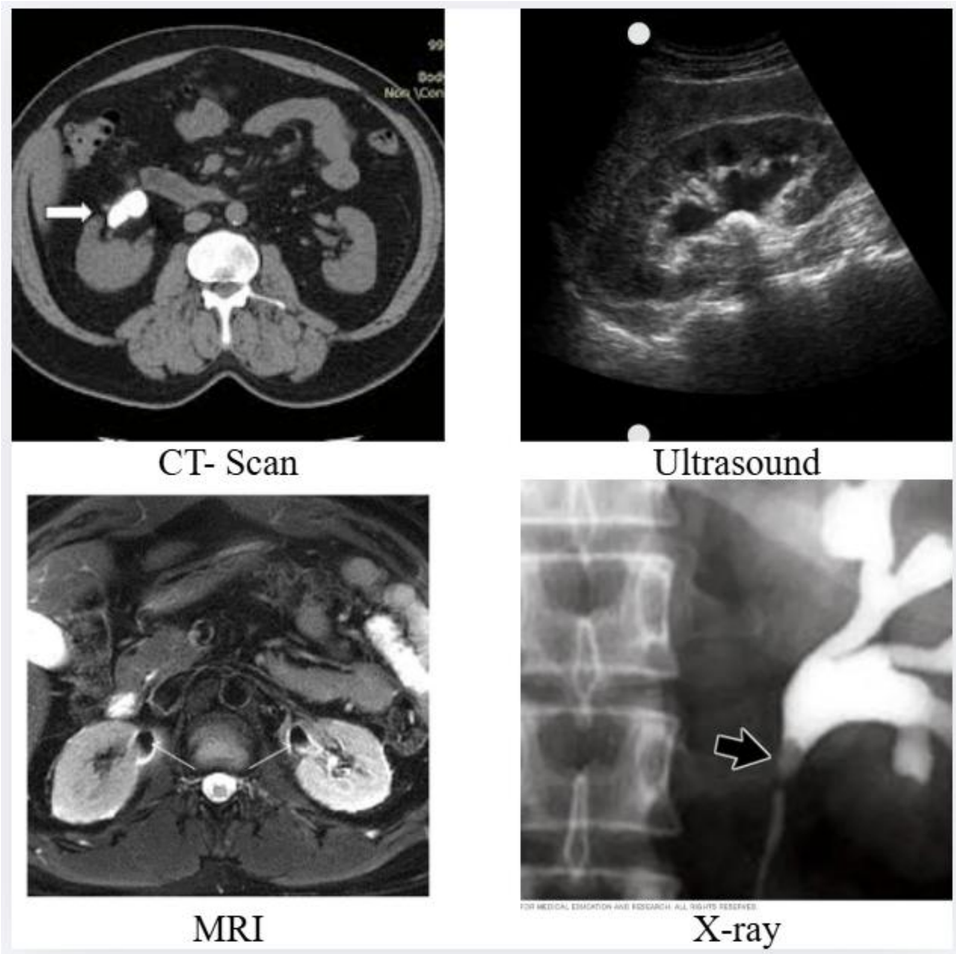


Figure 1.1 Comparison for Imaging Modalities for Kidney Stone Detection

The various imaging methods employed for the detection of kidney stones are depicted in Figure 1.1. The preferred method of imaging for kidney stones, used extensively in clinical practice, is ultrasound, owing to the fact that it is safe, cost-effective, and non-invasive. Therefore, this study examines ultrasound as a method of detecting kidney stones, and a number of studies focus on ultrasound-based detection methods.

Ultrasound imaging is becoming increasingly popular as a diagnostic tool because it does not involve any invasive procedures, is relatively inexpensive, and does not expose patients to ionizing radiation. Therefore, it has become a common method for identifying and monitoring kidney stones in clinical settings. There are many advantages to using ultrasound for diagnosis; however, ultrasound imaging also has a great number of challenges that impair the accuracy of the diagnostic images generated using ultrasound. For example, speckle noise and poor contrast between kidney stones and the surrounding tissue, as well as imaging artifacts such as acoustic shadowing, result in degraded quality of ultrasound images, making it difficult to accurately identify and localize renal stones (particularly small or low-density stones) [2],[5],[11].

Ultrasound Imaging presents one more challenge that affects ultrasound imaging is the differences between those conditions under which ultrasound images are taken. The variation can come from several things including variance between machines, type of frequency used by the probe, technique of the operator and the body composition of the patient will influence how well the image appears. This type of variability in data can result in inconsistent data, making it difficult for the clinician and software systems to provide an equal level of diagnostic accuracy and/or reliability from one clinical setting to the next. Consequently, it is imperative to develop imaging models that are robust against this variability and provide reliable detection across all types of clinical settings.

An additional limitation associated with diagnosis through ultrasound is that it depends on the expertise of the radiologist. Ultrasound image interpretation requires a high degree of expertise and skill, and trained operators may still experience difficulty distinguishing small features in the images. Consequently, different operators may arrive at different outcomes when interpreting images, resulting in inconsistencies and the possibility of missing some findings. Additionally, with the growing number of medical images produced annually as well as the increasing workload placed on radiologists, there is an increased chance of errors made by humans and also delays before a diagnosis is received. This further emphasizes the need for automated systems to help clinicians have consistent and unbiased diagnostic results [2], [3].

Recently, there has been great potential for AI/deep learning in tackling issues of medical images. For example, using Convolutional Neural Networks (CNNs) to classify/detect images through their automatic learning of complicated features from raw data provides a highly efficient means of executing these tasks. Recent studies have used CNNs for detecting kidney stones from ultrasound images with accurate and robust results [1], [5], [6], [17].

Thanks to object detection methods, automation in imaging has gotten even faster and more accurate due to the introduction of object localization (i.e., knowing where to find an object in the ultrasound image). Various models like YOLO (You Only Look Once) and its many derivatives are being used extensively for object detection, primarily because they have the speed and accuracy to perform real-time detection (YOLOv3, Faster R-CNN) [12],[14]. Each version of YOLO has continued to improve upon one another, with the latest versions of YOLO (YOLOv5 and YOLOv8) showing significant advancements over previous generations regarding identifying renal stones, speed, and accuracy of location [8],[19]. RT-DETR and Deformable DETR represent new advances in image recognition by capturing the overall context of the entire image [9],[18]. An additional approach for medical images is through vision transformers, which provide even greater improvements in detection accuracy for complex ultrasound images [23].

In addition to the progress made, there are still many obstacles that impact the creation of effective renal stone detection systems using deep learning technologies. One major issue is the lack of a sufficient amount of high-quality annotated medical datasets. Domain knowledge is necessary to annotate ultrasound images, and the time-consuming nature of this process leads to limited amounts of high-quality labelled data being available for training deep learning models. To help address this problem, more recent work has used semi-supervised learning methods such as pseudo-labels where a model assigns labels to unlabelled data, thus resulting in an increased amount of data available for training and better performance [4]. Moreover, data augmentation techniques also exist (e.g., GAN-based approaches) to increase dataset diversity and improve model generalisation [22].

A significant concern regarding the use of deep learning systems in medicine relates to a lack of interpretability. The majority of the models operate like black boxes, delivering predictions with no rationale behind the reasoning for the respective predictions made by the automatic system. Consequently, clinicians may have diminished confidence in using such automated systems due to this absence of rationale. In order to alleviate this issue, explainability techniques such as gradient-weighted class activation maps (Grad-CAM) have been developed to aid in the visualization of the area in the image that contributed to the model's decision and thus improve the transparency and assist with the clinical validation of model outputs [10].

Given these challenges, there is an urgent need for an effective, efficient method of identifying renal stones via ultrasound images. This study will develop a robust, interpretable model of renal stone identification by utilizing a deep learning architecture as well as applying new pre-processing techniques for improving ultrasound imaging, including denoising and enhancing contrast to obtain high-quality images, and creating additional labelled data via pseudo-labelling. The effectiveness of different models (Convolutional Neural Networks for classification and advanced object detection architectures) in both identifying and localizing renal stones will be evaluated.

Additionally, this proposed system will use techniques for better explanation of the model's predictions. These techniques are intended to provide an increased understanding of how the model arrived at its prediction so that it can be used more easily in clinical practice. The framework will also be able to be tested on both publicly-available datasets and real-world clinical ultrasound images so that the findings are valid across multiple populations. This applied deep learning approach will address practical considerations associated with the use of deep learning within a clinical environment and thus will contribute to the ultimate goal of creating an automated system that supports the healthcare provider in providing accurate, consistent and efficient diagnosis of kidney stones.

Healthcare and artificial intelligence are increasingly intertwined, emphasizing the need for intelligent diagnostic systems that assist clinicians in making accurate decisions. With the increased volume of complex data generated from medical imaging systems, it's becoming increasingly important to incorporate automated methods for interpreting this data. The use of deep learning techniques in ultrasound imaging represents a major milestone toward these goals. Automated systems will provide solutions to the issues of image quality, image variability and image interpretation that currently challenge traditional diagnostic methodologies. This study correlates with these developments by devising a feasible and clinically advantageous approach to detect renal stones and ultimately continues toward the overall agenda of data-driven healthcare.

1.2 Problem Statement

Even though there is widespread use of ultrasound in diagnosing renal stones, there still remains some ongoing challenges that prevent it from working well within a clinical setting. The inherent problems associated with ultrasound images are due to speckle noise, very low contrast between structures, and imaging artifacts like acoustic shadowing. All of these issues result in poorer quality images and make accurate identification and localization of renal lithiasis difficult; especially when dealing with a multitude of factors such as small stone size or low-density stones. Thus, this often presents a challenge for even the most experienced radiologists, who are at risk of missing or misinterpreting subtle findings.

Another key disadvantage of using ultrasound for diagnostics is that the interpretation of the images relies heavily on the interpreting clinician's education and experience. Therefore, there could be variances in how an individual clinician will interpret the images and perceive the patient based on a lack of objectivity in his/her evaluations, leading to different decisions made by different clinicians. As the complexity of the patient's anatomy increases or when multiple clinicians are involved with the patient's care over time, the variability will become even greater. These variances show the need for automated systems to support clinicians in providing more consistent and reliable outcomes.

In recent years, deep learning has brought about new advancements that have allowed researchers to create new techniques for improving the detection of kidney stones in ultrasound images; however, there are still significant limitations to these methods. One of the biggest challenges is the availability of medical data in the form of labelled annotations since the annotation process requires skilled professionals to do it properly. Because the annotations are an exhaustive task, they are very resource-intensive and time-consuming to collect, thus reducing the size of the training datasets available to use for modelling. This ultimately results in many AI-based models not being able to generalize well for new or unseen data.

A significant portion of current literature regarding the detection of kidney stones has focused on the classification of an image showing a kidney stone rather than on the actual location of the stone on an anatomical image. It is important to include localization information when studying kidney stones because it provides additional data about the size and location of the stone in an anatomical image. Object detection models (eg, YOLO, Faster R-CNN, transformer-based approaches) are developed to help localize kidney stones in noisy ultrasound images and define their exact location.

These models, however, still present major engineering difficulties in successfully localizing kidney stones in fuzzy ultrasound images.

Earlier, we stated that the current set of deep learning models have limitations in the manner in which they can generalize from their training data to clinical data. Most models are trained and evaluated using publicly available datasets that are typically 'cleaner' or more standardised than normal clinical images. Ultrasound images, however, are extremely variable based on the type of machine, the conditions under which they were captured (e.g. imaging settings) and patient conditions; therefore, a deep learning model trained on ultrasound images may not generalise well when exposed to actual clinical conditions.

The interpretability of deep learning in clinical environments presents many challenges. Generally, deep learning models operate as 'black boxes' so they provide predictions without explaining how they reached that conclusion. Because deep learning models provide no insights into their decisions, trust in their use by clinical providers is diminished and the use of automated systems in clinical workflows is limited. While methods such as Grad-CAM are available to render model attention visually available, their use in the delivery of patient care is still not widespread.

There is a significant need for a highly effective and powerful complete system to identify kidney stones from ultrasound images, overcoming the issues of noise and inconsistency in the data, as well as inadequate amounts of data to train on effectively. The system must be capable of detecting stones with high precision and will also provide an accurate location on the images as well as a consistent level of performance across multiple datasets, including pictures taken in a clinical setting. In addition, it must have a semi-automated mechanism for reducing the need for manual labelling (e.g., using pseudo-labelling) and offer an appropriate degree of explainability and transparency to develop trust with the clinical community.

The main issue being addressed in this study is creating a machine learning algorithm using deep learning techniques to analyse images collected from ultrasound scans to locate kidney stones. The goal of this project is to create an automated (no human input), dependable, and interpretable machine learning based algorithm as an alternative to traditional diagnostic methods and existing computational methods. This will involve developing and integrating state-of-the-art deep learning-based detection models, data augmentation (enhancing training data), and explainability techniques into a single framework. This will provide a clinically useful diagnostic method of detecting kidney stones in ultrasound images while also scalable to meet clinical demands.

Identifying the aforementioned limitations show that traditional methods cannot adequately resolve the intricate problems associated with the detection of kidney stones in ultrasonographic images and that a valid comprehensive solution must include accuracy, strength, scalability and interpretability. The development of a technical solution that meets all four parameters, thus creating an effective as well as clinically relevant solution, is contingent upon the identification and subsequent resolution of the interconnected concerns addressed in this study whereby the basis of the problem defined in this study leads to the motivation for the deep learning solution proposed herein.

1.3 Need of the Study

Due to the rising incidence of kidney stones worldwide, healthcare systems view it as a major concern. The numbers of people with the condition have grown steadily worldwide due to dietary changes, less physical activity, dehydration, and environmental factors, among others. Besides the decreasing incidence of kidney stones, they can recur in many of the same patients; therefore, this adds to the burden of kidney stones on patients and healthcare providers. Given these circumstances, there is an urgent need for effective testing to facilitate early diagnosis and reduce the incidence of complications [1], [3].

Ultrasound imaging is commonly used to diagnose kidney stones due to its ability to obtain images without surgery and at a low cost. However, there are several limitations related to the use of ultrasound imaging for this purpose. First, the presence of speckle noise and low contrast between the stones and surrounding soft tissue degrades the quality of ultrasound images and therefore makes it difficult to obtain accurate images of renal stones, particularly with small or low-density stones. Additionally, ultrasound-based diagnoses are highly reliant on the radiologist's level of expertise, which can lead to differences between interpretations and potentially different results [2], [5], [11].

As the amount of medical imaging data being generated is increasing at an unprecedented rate, radiologists have had to process more images than ever before, resulting in increased workloads and the potential for human error and delays in diagnosis. In these situations, automated computer-aided diagnosis (CAD) systems would be able to provide radiologists with the ability to make faster and more accurate decisions with the same degree of consistency across multiple diagnoses [2], [3].

Deep learning has recently made great strides toward creating automated systems for analyzing medical images. With the help of deep learning, models have been able to extract significant features from complex images and accurately conduct various other types of analysis, including classifying and detecting objects. However, many of these deep learning models have been trained and tested on a small number of restricted datasets, making it difficult for them to generalize to many real-world situations. Thus, there is an obvious need for the development of robust models, which exhibit the same level of performance regardless of the variety and noise level of the ultrasound data. [1], [5].

One of the primary challenges in this area is the lack of good quality labelled datasets. Annotating medical images takes a lot of time and resources and requires specific domain knowledge. Therefore, many studies are conducted with small datasets, and this has an impact on the ability of the models to perform and scale. Approaches such as pseudo-labelling and data augmentation have been developed to use unlabelled data to increase both the size and diversity of training data, thereby improving the robustness of deep learning models [4],[22].

Aside from the accuracy of detection, localizing the position of renal stones on ultrasound images is much more clinically relevant than just knowing that you have kidney stones because of classification models. Classification models will tell you that you have kidney stones but will not tell you the exact location of where they are located; whereas, by using an object detection model, there are bounding box predictions provided by the model to determine where the stone is located as well as its size. Knowing this information is critical for making clinical decisions and developing a treatment plan; therefore, a focus on detection-only methods that will provide both classification and localization (bounding box) is important [8], [14].

A crucial element that requires focus on is how to explain the decisions made by deep learning algorithms. While issuing accurate results is essential in a healthcare context, it must also be understood how they are made to help create trust between doctors who will use the algorithm once they have gone through clinical validation testing in real life. Deep Learning systems are sometimes said to be non-transparent and result in lowering a clinician's willingness to trust the output and therefore will not use the model. Grad-CAM is one of many methods available to provide a graphic representation of areas within an image that are relevant in computing predictive outcomes thus increasing the transparency of Deep Learning systems and helping to support Clinical Validation.

Additionally, a significant division exists between research-based models and their actual use in clinical settings. Numerous studies show great success when controlled but do not perform as well with real-world clinical data. These variances occur as a result of differences in imaging devices, acquisition protocols, and patient attributes. To bridging this gap, it is necessary to create systems trained and tested using both public datasets and real-world clinical datasets providing them with credibility for real-life use [2], [3].

Taking into account the above elements, there is a major requirement to create an improved, dependable, and easy to understand capability for renal calculi detection with ultrasound images. This study will address those basic challenges by developing a comprehensive and deep learning model enabled framework combining preprocessing methodologies, pseudo-labelling methods, and current detection systems. Also concentrating on both accuracy and understanding, and subsequently validating within multiple datasets, this research provides an implementing and supporting tool for clinicians to achieve quick and accurate diagnosis of renal stones.

The growing need for accessible medical services in areas that lack resources is another important factor that demonstrates why this research is necessary. Advanced imaging technologies, such as CT scans, are too expensive and typically not available in most areas, especially in rural or underdeveloped areas. The use of ultrasound is sometimes the only way to provide diagnostic images in these areas because ultrasound machines are relatively affordable and portable compared to other imaging devices. By using automated systems to improve the accuracy and reliability of diagnoses made using ultrasound, we can help to improve accessibility for medical care and provide treatment in a timely manner.

The reason for this project comes about because of the challenges associated with rising rates of disease, limitations of currently used tools to diagnose and find ways to treat diseases, and the increasing demand for smart healthcare solutions. This research proposes to develop a system to use 21st century deep learning tools along with realistic aspects like variability of data, and how 'explainable' the data, and the system will become with the use of deep learning. Ultimately, this work will produce a tool that will provide more accurate diagnoses, or at least provide a very good basis for such diagnoses, and be applicable for real-life use in clinical settings.

1.4 Objectives

This research project aims to build a solid and trustworthy deep learning driven platform for finding and pinpointing kidney stones by using ultrasound imaging. This research will deal with obstacles like background noise, low contrast, dataset limitations, and interpretability; while improving performance and generalisation in real-world clinical environments.

- Developed an automated detection System of Kidney Stones from Ultrasound Images
- Created a deep learning-based framework which classify and localize the presence of kidney stones within ultrasound images.
- Implemented and evaluated several Deep learning models to detect kidney stones.
- Compared the performance of object detection versus classification model performance.

- By using pseudo labelling techniques, improved the dataset quality through expanded labelling.
- Improved the model's ability to handle noisy/low contrast ultrasound images through increased robustness.
- Used image enhancement techniques including median filtering and CLAHE to preprocess the ultrasound images.
- Measured the performance of the models using common performance metrics like accuracy, precision, recall and mean Average Precision(mAP).

- Determine the model's generalization capability using both public and real clinical datasets.
- Use explainability techniques, e.g., Grad-CAM, to help interpret the models' predictions.
- Create an end-to-end pipeline solution for the detection and analysis of kidney stones.
- Provide the end user with a user-friendly system, so that it may have real-world clinical applications.

1.5 Scope and Limitations

1.5.1 Scope of the study

This study focuses on the creation and assessment of a framework that uses deep learning for detecting and localizing kidney stones in ultrasound images. The goal of this research is to create an automated system capable of accurately processing ultrasound data, detecting the presence of kidney stones, and locating them in the image. The study will be performed using a variety of different criteria including datasets, methodologies, model development, and their actual application.

The research collected both publicly accessible ultrasound datasets and clinically acquired ultrasound images to conduct an initial evaluation of model performance in a lab setting using publicly available ultrasound datasets, then evaluate those same models under real-world, clinical conditions using clinically acquired data. Both sources of ultrasound data will be evaluated using an identical framework to improve the overall robustness and generalizability of the proposed framework through a dual-source data analysis method. In addition, the data set has been divided into a training and validation set to facilitate an objective evaluation of the model performance.

Methodology: The research will use pre-processing methods that enhance the quality of ultrasound images, such as noise reduction using median filtration and enhancement of contrast using CLAHE (Contrast Limited Adaptive Histogram Equalization). The study will also implement data augmentation methods to increase the variability within the dataset so the model will perform better at generalizing under various imaging conditions. Lastly, the researchers will use pseudo-labelling methods to create a larger labelled dataset with no additional manual annotations, ensuring the continuity of the dataset.

Both classification and object detection approaches will be included in the model development process. Classification methods that make use of CNNs (e.g., ResNet and EfficientNet) will be addressed; object localization methods made use of advanced detection models (e.g., YOLO and RT-DETR) will also be included in the project. All models will undergo rigorous training and evaluation by virtue of being subjected to uniform testing conditions in order to produce statistically comparable performance results among the various types of classifiers and detectors.

The purpose of this system is to be able to detect the presence of renal stones as well as determine their location. Traditional classification procedures only provide an indication of whether or not there are stones present; however, this method provides bounding boxes to predict where the stones are located in the ultrasound images. The ability to localize the stones is important in clinical use for determining the size and location of the stones.

The other major part of the research involved using Explainable AI approaches. One way we can do this is by using a method called Grad-CAM which can help visualize parts of an image that were involved in decision making by the model. Being able to interpret a model will also help with its integration into how we use it in a clinical setting.

The study's application limits the use of the application to aid healthcare providers when giving kidney stones diagnostic information. The system is designed as a tool that will help healthcare professionals make decisions about patients with kidney stones but is not intended to replace medical specialists. Rather, this system provides added information about kidney stone diagnosis to radiologists to increase diagnostic accuracy and improve efficiency in performing diagnosis of kidney stones.

The last experimental aspect of this work consists of a comparative analysis that assesses the effectiveness of each renal stone detection model based on performance measurements such as Precision, Recall, and mean Average Precision (mAP).

In addition to addressing current requirements, this study considers future scalability and adaptability of the proposed system. Because deep learning is advancing rapidly, the framework developed during this study will continue to evolve beyond its initial constructs and will allow for the integration of new architectures and optimization methods. Therefore, if the system is adaptable, it will continue to be relevant over long periods, based on the availability of greater amounts of data and more powerful computers. Flexibility is an integral feature of successful transitions from research-based to clinical application models.

The research will also examine whether or not the new system can be adapted to different health care facilities. The system's framework is built with modular components so that enhancements can be made in the future if new datasets or improved machine learning architectures become available. As a result, the system will continue to be viable when new technologies are created. The project not only considers developing online tools in which clinicians can upload ultrasound scans to receive prediction and visual explanation but also the project's proposed approach has the potential to have widespread utility beyond just an experimental setting.

1.5.2 Limitations

Even with a well-developed model, the study has inherent constraints that need to be considered. The model is limited by the amount of data available to develop an algorithm for the basis of deep learning, which is the use of human-generated data (i.e., ultrasound images). Even though the model uses real-world ultrasound images, the size of the dataset is small compared to that required for large-deep-learning models and as a result will not generalise effectively to a wide range of clinical situations.

Additionally, an imbalance between positive and negative classes in the data set can result in fewer negative class instances. Fewer examples of a negative class may prevent the training model from developing the skill of accurately discriminating between abnormal and normal instances; this lack of skill may reduce overall system accuracy.

There is also a natural variance in the images produced due to differences between machines, methods of capturing and imaging, and a person's skill at imaging the patient, which may contribute to variability in images produced by ultrasound and complications related to the training and testing of a particular model. Therefore, the model developed will perform differently when using images from different clinics.

The result is that the study uses annotators' labels (manual labelling and pseudo labelling) to generate labels for all of the data. Although pseudo labelling can add to the number of labelled data items to train the model, any errors will result in poor quality training data if the initial model's predictions are not correct, which ultimately affects the performance of the model.

Another drawback of the work presented is that the developed system was only tested in a laboratory setting, and has not yet been implemented in an actual clinical environment, as a result of which aspects of computational efficiency and integration with the hospital system, and the computational and processing requirements of a real-time system, were not fully addressed by this research.

The other major part of the research involved using Explainable AI approaches. One way we can do this is by using a method called Grad-CAM which can help visualize parts of an image that were involved in decision making by the model. Being able to interpret a model will also help with its integration into how we use it in a clinical setting.

In conclusion, although many practitioners have incorporated explainability techniques as a means of making their models more interpretable, these techniques do not always provide a full representation of the decision-making process of deep learning models. As such, it is recommended that the system be monitored by clinicians to facilitate the correct diagnosis.

The type of data used to train deep learning models can have a huge impact on their performance, especially in regards to how good or diverse that data is. Therefore, even small amount of bias in a dataset will affect what type of inferences can be drawn from the model over its entire range of performance and may make it difficult to apply the model across multiple demographics. Also, due to the differences in ultrasound imaging protocols within institutions, tuning the model prior to implementation may also be necessary. Furthermore, the type computational resources required to train each advanced deep learning model can be a limiting factor, particularly for facilities with limited computational capacity. In order to address all of these limitations, additional research, larger datasets and clinical expert partnerships will be required to develop a solution that is sufficiently general and deployable across a wide variety of applications.

Chapter-2 Literature Review

This chapter reviews existing work on renal stone detection in ultrasound images. It includes a detailed literature review, a comparative analysis of key studies, and identification of research gaps.

2.1 Review of Existing Work

Advancements in the use of artificial intelligence in medical imaging for analysing and interpreting ultrasound data. R. Jha et al. [2] discuss how AI could provide us with opportunities and challenges with handling noisy, low-contrast ultrasound imaging data. In the same vein, S. Liu et al. [39] provided a comprehensive overview of the use of deep learning techniques in the feature extraction and pattern recognition of ultrasound images.

Deep learning has emerged as a strong mechanism to process medical images, most significantly via Convolutional Neural Networks. R. Yamashita et al. [48] provided an overview of such CNN based approaches in the field of medical imaging, and they demonstrated the ability of CNNs to automatically learn complex features from images through a process called feature extraction. Further, D. Shen et al. [50] provided various methods of performing medical image analysis using deep learning techniques and indicated that these methods can improve the accuracy of detection as well as reduce the need for manual intervention with regard to medical image analysis.

Prior to deep learning techniques being adopted, standard methods of using ML were widely used in ultrasound imaging for detecting kidney stones. For instance, texture-based feature extraction methods followed with the classification of features using SVM were shown to be able to accurately identify kidney stones based on M. Hassan et al.'s [11] evaluations of this method; however, they could only be used if appropriately designed and required the use of domain knowledge.

Advancements continue with blending feature-based techniques and machine learning algorithms. For example, R. Ravi et al. [33] used feature extraction to create a model for kidney stone detection, which showed moderate success but had limited ability to adapt to differences in quality of ultrasound images in different patients. These methods have found it difficult to generalize to multiple datasets that have different types of noise and/or imaging conditions.

Recent studies have focused on enhancing the ability of machine learning to detect objects by improving the extraction of features and classifying them into categories. For example, Y. A. Asaye et al. [24] developed a machine learning-based method of detecting kidney stones using ultrasound images that outperforms previously developed techniques in terms of success rate. However, both techniques are based on handcrafted features, which limit their ability to be made into a machine learning model that can be used in practical clinical applications due to the need for significant time spent on each feature.

As deep learning advanced, the use of convolutional neural networks (CNNs) in the field of medical imaging started to replace more traditional forms of machine learning. A study by A. Chouhan et al. [6] looked at using CNNs to classify kidney stones in ultrasound images. The results of this study showed that using CNNs outperformed more traditional approaches that used features based on expert knowledge. CNNs can automatically learn to extract important features from raw images rather than requiring the manual extraction of features by a medical professional.

Using data augmentation or deeper architectures to extend the performance of CNN-based methods led to greater improvements than their counterparts. S. Singh et al. [5] have shown that combining data augmentation with Deep CNN models improves the robustness of their model as well as the accuracy of its classification results. A similar conclusion was reached by Y. Ma et al. [3] when they developed a CNN-based technique for detecting renal calculi; this demonstrated that Deep Learning models are capable of extracting complex relationships/structures from ultrasound images.

Transfer learning methods were developed to overcome the limitations associated with dataset size and the generalization of models. For example, Y. Sun et al. [13] have shown that transfer learning yields improvements in performance when training on small ultrasound datasets by applying transfer learning to the detection of kidney stones. A. Banerjee et al. [17] have shown that using a pre-trained deep-learning architecture such as VGG16 can provide a better accuracy in classifying renal stones compared to using a standard approach.

To increase the effectiveness of CNN-style systems an introduction of hybrid architectures that combine various techniques is one way of doing so. The CapsuleNet-CNN hybrid model proposed by T. Ahmed et al.[15] presented a combination of CNN-based feature extraction and classification capabilities for renal ultrasound images leading to improved classification accuracy; enhanced ability to represent features and greater representation accuracy through improved representation of spatial relationships between objects in their original position within the ultrasound images due to the combination of capabilities afforded by the convolutional layers in addition to that of the capsule networks.

There have been attention mechanisms added to convolutional neural networks (CNNs) as well, which has helped to improve the extraction of features through a CNN by concentrating on relevant areas within the image. L. Tao et al. [20] developed a CNN with an attention mechanism to classify renal ultrasounds and improved classification accuracy by placing more emphasis on the important areas of the image. Similarly, X. Chen et al. [40] studied the use of attention mechanisms for deep learning-based medical image segmentation and detection applications and found that their use greatly improved the performance of the model for more complex imaging tasks.

To enhance the robustness and reliability of kidney stone detection, ensemble learning techniques have been implemented. For example, R. Zhao's team [21] used an ensemble of CNNs to classify ultrasound images, yielding improved accuracy using the combination of multiple models' predictions. As a result of utilizing ensemble models, the model's variance was reduced and generalization performed better on all samples in the dataset.

Techniques for increasing sample numbers via augmentation for reliable learning has significantly developed ML performance. Example of this is found in the works by R. Patel et al. [22]. Augmentations with GANs were created and used in order to generate additional training samples because of the limited size of the anatomy datasets. As a result of using augmentation methods, the robustness of ML models improved. Furthermore, the employment of augmented samples in learning improved how well the resulting models handled the natural variances present in the ultrasound images.

Techniques for detecting objects in pictures have become very popular with the goal of finding and pinpointing areas in pictures that are of interest to doctors. The first methods developed used pixel-by-pixel predictions to accomplish this goal and included the use of fully convolutional networks. By using a fully convolutional network for semantic segmentation, J. Long et al [49] created the first end-to-end method for producing dense predictions and creating the basis for modern detection and segmentation models.

The performance of object detection has been enhanced by the addition of region proposal algorithms in region-based Convolutional Neural Networks (RCNNs). For example, by applying Faster RCNNs to detect kidney stones in ultrasound images, K. Ueda et al. [14] have shown that localizing kidney stones with an RCNN achieves a significantly higher level of accuracy than typical classification techniques. These models can be used for automated identification of object boundaries in the context of automated medical detection processes.

Subsequent to the earlier developments, these Single-stage object detection models are able to optimise the speed of object detection while also maintaining a level of accuracy that is competitive with other methods. Z. Li et al. [12] showed that it was possible to perform real-time detection when using YOLOv3 for the detection of ultrasound lesions from images; and J. Park et al. [16] used RetinaNet to detect lesions within ultrasound images, while benefiting from a focal loss, as a means of dealing with the inherent imbalanced nature of medical datasets.

Advanced technologies have recently developed using transformer-based detection architectures, resulting in improved performance for detecting objects in complex scenes. For example, Y. Zhang et al. [18] created deformable DETR for ultrasound object detection with a clear improvement in detection through the use of attention mechanisms. These types of models also provide improved capabilities for capturing global contextual information and accommodating changes to the size/shape of objects being detected.

The YOLO Model Family is known for being one of the fastest and efficient real-time object detection models. J Redmon et al. [41] have made several advancements (YOLOv3) providing researchers having adequate performance when detecting objects within one frame Photographs. Brides from signal detection frameworks to unified detection framework design, such as multimedia patient.

Further developments of the YOLO design were focused on improving the performance accuracy and speed of the model. A. Bochkovskiy et al. [42] created YOLO version 4, which has many upgraded training methods and architectural changes, resulting in improved detection rates. These improvements allow YOLO-based models to perform complex detection operations with more reliability than previous versions of the YOLO architecture.

The YOLO Changes to Increase the Similarity between YOLO Products and Algorithms to Make Detection More Efficient and Effective. Z. Ge at al. [44] created a development called YOLOX. YOLO had improvements in various areas, including the YOLO Training Strategy and the YOLO Model Design, therefore, improving the Accuracy of the both YOLO and your computer (for Real Time Speed). This Change has Strengthened the Ability of the YOLO's Models for Use in Medical Imaging.

The newer YOLO versions have progressively enhanced their ability to detect objects via improvements to the architecture used by YOLO, culminating in YOLOv7 as explained by C. Wang et al. [43]. YOLOv7 set the current state-of-the-art for real-time detection of objects. V. Nguyen et al. [28] also added attention mechanisms to YOLOv7, which enhanced the ultrasound image feature extraction and detection accuracy.

Many researchers have used YOLO algorithms and models for detecting and identifying kidney stones in medical imaging. V. Reddy et al. [19] article used YOLOv5 to perform real time detection of kidney stones in ultrasound imaging, showing the ability of YoLOv5 to identify and locate kidney stones. Additionally, these models present a good balance between speed and accuracy for use in clinical practice.

Recent studies on the use of advanced YOLO models for renal (kidney) stone detection have concentrated on utilizing YOLO v8 models. K. Murthy et al. [8] used a YOLO v8 model for detecting kidney stones in ultrasound images and reported enhanced performance compared to previous versions of the YOLO model. N.C. Sai et al. [25] created YOLO v8 based detection frameworks for kidney stones and also demonstrated the effectiveness of the model for use in other medical imaging applications.

Many authors have proposed changes to YOLOv8 that could improve its performance. The work by H. Huang et al. [26] introduced LG-YOLOv8, which is an enhanced version of YOLOv8 that improves the accuracy of detection via refinements in architecture. These enhancements provide better results when working with ultrasound images, including those that have noise and variability.

YOLO-based models remain challenged by factors that affect accuracy of detection within medical imaging scenarios. Many of these challenges arise from the presence of noise in the image being captured (e.g., noise due to the patient's body), low contrast between the target and the background, and variability associated with the appearance of stones within human bodies [2], [39]. Furthermore, significant advancements have been made in YOLO architectures and method of training on data resulting in more robust models for real world clinical applications [41], [43].

The evolution from YOLOv3 to YOLOv8 demonstrates that there have been major advancements in the precision of detecting objects [41], [42], [43]. As a result, these models have emerged as optimal solutions for real-time medical image analysis relating to accurate localizing and computing efficiently, making them ideal for automatically detecting renal stones [8], [25].

Deep learning has developed in the past few years, and now the Transformer-based architecture is developed for the purpose of medical image analysis, allowing us to obtain full global context. W. Huang et al. [23] have used vision transformers to analyse ultrasound images, illustrating that they can represent long-distance dependencies and that vision transformers can provide a better representation of the features than convolutional approaches.

Various forms of computer vision tasks utilize transformer models with attention-based mechanisms to yield good results. For instance, H. Touvron et al. [47] introduced Vision Transformers for large-scale image classification and showed that they effectively capture features globally. Therefore, vision transformers have been very popular in recent years for improving detection and classification performance in medical imagery.

Transformer-based models can offer many advantages; however, they typically require many data points and a significant amount of processing power. This can restrict how they can be used in situations where there is limited data available (such as in many images found within a medical folder). Because of this restriction, the research community continues to investigate how to integrate transformer architectures into an existing convolution neural network (CNN) [23], [47].

In order to increase both accuracy and reliability of detection, multi-stage and hybrid detection frameworks have been explored in the literature. R. Rao et al. [27] provided a two-stage deep learning approach to detect kidney stones and assess their severity by using both YOLOv8 and ResNet-18. The use of both models allowed for better localization and classification of detected kidney stones than would be possible with one solitary detection model; thus, improving the overall performance of kidney stone detection as a whole.

Previous tests with hybrid detection models, in addition to multi-stage detection approaches have been used to improve detection efficiency. P. Kumar et al. [7] used the EfficientDet model to improve the accuracy of medical ultrasound imaging stone detection by providing a better balance between detection performance and computational efficiency through efficient feature scaling and optimized architectural design.

Overall, multi-stage and hybrid detection systems are able to provide an improved level of performance by combining the advantages of different architectural systems. At the same time, they may also create additional complexity and computational load, thereby making them less suitable to operational use in a real-time clinical facility.

A major challenge in the distribution of high-quality labelled datasets in the medical field is the limited availability of them. To overcome this issue, researchers are investigating several different methods including semi-supervised learning approaches. In a paper by S. Mehta et al. [4], a pseudo-labelling-based framework for renal ultrasound analysis was developed. The authors demonstrated that unlabelled data could be used as a source to enhance the performance of models created from labelled data alone.

Detection of ultrasound images is hindered by various factors including noise and low contrast. S. Obaid et al. [29] evaluated the classification of kidney ultrasound images that were corrupted by noise with the help of ensemble deep learning (EDL) models. The results indicated that the use of EDLs provided greater robustness to noisy ultrasound images compared to single model EDL approaches.

One of the critical elements of enhancing performance of a model relates to the choice of relevant frames from the ultrasound data collected during a procedure. The work performed by A. Seraj et al. [30] examined an automated frame selection method that allows for the extraction of high-quality frames from kidney ultrasound videos to be used for training deep learning models (improved quality of data used to train the model and increase the performance of the trained model).

The selection of frame; characteristics of the dataset itself (design/sizing); and variability/number of images in each dataset can impact the deep learning model's ability to generalise from training set examples. M. Panchal et al [31] investigated the efficacy of using different medical imaging datasets for detecting kidney stones in a deep-learning context and illustrated how having a wide range of examples from different datasets is critical for consistent accuracy on the models developed.

Significant remaining challenges in ultrasound image capture are imbalance of datasets, noise, and variability. These elements can impact model training and decrease the accuracy of detection of data collected in clinical settings [2], [39].

The limitations described have led several different methods to be employed, for example, data augmentation, pseudo-labelling, and ensemble learning techniques. These approaches can support greater robustness within models, improved ability to generalise from their training data, and perform better under many different conditions of ultrasound imaging [4], [29] and [31].

With the increasing complexity of deep learning models comes an increased demand for interpretable options within the field of medical imaging. Explainability techniques assist with model interpretation and help to foster trust among users who utilize the models. D. Das et al. [10] demonstrated the use of Grad-CAM for classifying renal imaging ultrasound (US) images and showed how visual explanations can be used to indicate the location of critical areas of interest in US images.

The "Gradient Class Activation Mapping" (Grad-CAM) is one of the best-known explanatory techniques in the field of deep learning. As described by R. R. Selvaraju et al. [38], Grad-CAM creates a heat map that shows how different parts of a given image influenced the prediction made by the neural network. This improves the overall interpretability of a deep learning model without changing the underlying architecture.

Techniques for providing explanations have been studied in order to create greater transparency in medical imaging systems. In their paper, S. Rajaraman et al. [34] explain that visual explanations are critical to validating the decision-making of models and are therefore important in creating trust in their reliability.

Explainability is very important for identifying anatomically relevant areas of focus when detecting kidney stones with a computer-based model. Using visual methods to provide explanations for a model’s predictions (e.g., using Grad-CAM heatmaps) can help clinicians evaluate the reliability of the model's output in clinical settings, and ultimately improve the clinician's ability to interpret the model output.

Even though there are some positive aspects from using various techniques for generating explanations, there could still be challenges with the accurate and consistent interpretation of the results. For example, heat maps created by these explanation techniques can sometimes provide evidence for regions on the map to be interpreted as irrelevant, particularly with noisy ultrasound images. Thus, it is important to evaluate the results of the explanation prior to developing a meaningful interpretation of the results [10], [34].

The use of explainability approaches such as Grad-CAM in modern medical imaging systems plays an important role; they increase transparency, improve user trust and help to facilitate clinical decision making by providing visual information about model predictions [36], [35].

Deep learning architectures are key components to improving tasks related to analysing medical images. Convolutional Neural Networks (CNNs) can be used as the backbone for different detection and classification models. K. He et al. [35] created the ResNet architectures that utilize residual connection to help assist the problem of vanishing gradient by being able to train deeper layers of a neural network.

Medical imaging has seen an increased use of ResNet-based architectures as they can learn complex hierarchical features from images. Not only have these models achieved or excelled at classification tasks, but they've also been used as part of hybrid models for detection and localization tasks as well.

Recent improvements in deep learning architectures have been geared towards greater model efficiency and scalability. M. Tan et al. [36] proposed EfficientNet, which is based on a compound scaling approach that balances the depth of the model, the width of the model, and the resolution of the image. As a result, they were able to achieve superior performance at lower costs to run the model, as compared to the practices for scaling existing networks previously used.

Models based on EfficientNet have found extensive usage in the field of medical images because they give higher accuracy while requiring fewer parameters. Their architectures are particularly well adapted for computing resources, making them a good fit for both practical and real-time applications in healthcare.

The loss function is essential in guiding the optimization of object detection models during training. When performing object detection in the medical imaging domain, background and target regions usually have unequal class frequencies. Therefore, the selection of an appropriate loss function is key to improving detection performance.

Focal loss, developed by T. Y. Lin et al. [45], is an alternative to the traditional cross entropy loss function that solves the problem of class imbalance, such as the high number of easy-to-detect negative samples when training using cross entropy, by decreasing the weight or contribution of those samples; therefore, focusing the model's attention on hard-to-detect samples, thus improving the accuracy of detection on difficult-to-detect cases.

Object detection models frequently utilize focal loss, especially in areas such as medical imaging, where small objects like kidney stones are difficult to detect accurately. With a focus on difficult-to-detect examples, focal loss allows for improved model sensitivity and the reduction of false negative detections.

In the past few years, several researchers have been investigating alternative imaging techniques for identifying kidney stones other than ultrasound. For example, a recent study by Y. Tang et al. [32] used computed tomography (CT) images of kidney stones, which enable the clear anatomical features of the stone, to develop a deep learning program for accurately detecting stones. As a result, the accuracy of this CT-based detection system is significantly higher than that of ultrasound-based detection systems due to the greater clarity of the source images.

While CT imaging is accurate in detecting kidney stones, cost considerations and radiation exposure limit its routine use in clinical practice. In contrast, ultrasound imaging is safer and more readily available, making it better suited for the real-time and non-invasive evaluation of kidney stones.

2.2 Comparative Analysis Table

In Table 2.1, comparative analysis of existing methods that have been explored during the study of renal stone detection is shown.

Table 2.1: Comparative Analysis of Existing Methods for Renal Stone Detection

Ref & Authors	Data Source	Methodology	Key Metrics & Results	Contribution	Limitation
[1] Nowak et al., 2021	Public ultrasound dataset	Deep CNN	Accuracy: ~0.88, Precision: ~0.86	Feasibility of deep learning	Limited dataset
[3] Ma et al., 2024	Public ultrasound dataset	CNN detection	Accuracy: ~0.90, Recall: ~0.88	Improved detection	No localization
[4] Mehta et al., 2023	Public + unlabeled ultrasound	Pseudo-labeling	Accuracy: ~0.88, Recall: ~0.87	Reduced labeling effort	Error propagation
[5] Singh et al., 2022	Public ultrasound dataset	CNN + Augmentation	Accuracy: ~0.92, Precision: ~0.91	Better generalization	Classification only
[6] Chouhan et al., 2019	Public ultrasound dataset	CNN	Accuracy: ~0.85	Better than ML	No localization
[7] Kumar et al., 2023	Public ultrasound dataset	EfficientDet	mAP: ~0.78, Precision: ~0.80	Balanced performance	Needs tuning

[8] Murthy et al., 2024	Clinical ultrasound dataset	YOLOv8	Precision: ~0.92, Recall: ~0.95, mAP: ~0.90	Real-time detection	Small object issue
[9] Lin et al., 2023	Public ultrasound dataset	RT-DETR	mAP: ~0.75, Precision: ~0.78	Global features	Needs large data
[10] Das et al., 2024	Ultrasound dataset	Grad-CAM	Qualitative heatmaps	Interpretability	Limited depth
[11] Hassan et al., 2020	Public ultrasound dataset	SVM + Texture	Accuracy: ~0.75	Early approach	Feature dependency
[12] Li & Wang, 2020	Public ultrasound dataset	YOLOv3	mAP: ~0.70, Precision: ~0.72	Real-time	Lower accuracy
[13] Sun et al., 2021	Public ultrasound dataset	Transfer Learning CNN	Accuracy: ~0.89, Precision: ~0.87	Pre-trained benefit	No localization
[14] Ueda et al., 2021	Clinical ultrasound dataset	Faster R-CNN	mAP: ~0.80, Recall: ~0.82	Accurate detection	High cost
[15] Ahmed et al., 2021	Public ultrasound dataset	CapsuleNet + CNN	Accuracy: ~0.91	Spatial learning	Complex

[16] Park et al., 2021	Ultrasound dataset	RetinaNet	mAP: ~0.76	Effective detection	Not real-time
[17] Banerjee et al., 2022	Public ultrasound dataset	VGG16	Accuracy: ~0.90	Strong baseline	No localization
[18] Zhang & Luo, 2022	Public ultrasound dataset	Deformable DETR	mAP: ~0.78	Region focus	High compute
[19] Reddy et al., 2022	Ultrasound dataset	YOLOv5	mAP: ~0.82, Precision: ~0.84	Fast detection	Sensitive
[20] Tao et al., 2023	Ultrasound dataset	Attention CNN	Accuracy: ~0.91, Recall: ~0.89	Feature focus	No localization
[21] Zhao et al., 2023	Ultrasound dataset	Ensemble CNN	Accuracy: ~0.93	Robust	Complex
[22] Patel & Bhatnagar, 2023	Public ultrasound dataset	GAN augmentation	Accuracy gain: ~0.89	Data diversity	Unrealistic risk
[23] Huang et al., 2024	Large ultrasound dataset	Vision Transformer	Accuracy: ~0.92, Precision: ~0.90	Global features	Needs large data

2.3 Research Gaps Identified

In this section, we examine the main limitations found in previous studies: traditional and deep learning limitations regarding object detection performance; the available datasets; the quality of the datasets; generalization of the models; and finally, the explainability and integration of the complete clinical system.

Traditional and early deep learning (DL) approaches to ultrasound imaging of kidney stones are limited. These techniques were based heavily upon feature extraction (machine learning). Feature extraction used hand-crafted features and therefore could not be applied directly to data collected under varying conditions. As such, while they may have performed well in a controlled environment, their overall generalizability across multiple datasets is limited.

Convolutional neural networks made it possible to automatically discover relevant, discriminative features without requiring human-defined rules (manual feature extraction). These new deep learning models have been shown to provide better classification accuracies than traditional machine learning techniques. The drawback is that they still had challenges dealing with the variations and noise associated with ultrasound images, which limited their usage.

Superior performance was obtained with both the use of deeper architectures and by utilizing techniques for transfer learning. However, in order for these models to have good performance, they generally require large and diverse source datasets. For instance, the performance of those models will often suffer when applied to new or clinically dissimilar data, which represents a challenge associated with producing consistent generalization.

Traditional methods of machine learning and deep learning generally were used as methods of identifying image classifications in the medical field; however, they did not consider the location of the kidney stones, but only the fact that they existed. In medical settings the exact location of the kidney stones must also be identified when making treatment and diagnosis decisions; therefore, there was a limitation to the use of the classification approaches in a practical setting so detection-based methodology is now needed to perform more effectively.

This part of the paper addresses some limitations of object detection models, specifically those based on YOLO, when used for medical images. Even though there has been tremendous progress in terms of detecting kidney stones and localizing them, these models have much less success with ultrasound images than we've seen in other imaging modalities. The presence of noise in images, low contrast and variability in stone appearance all lead to decreased accuracy in detecting stones and diminished confidence when applying findings in a clinical setting.

Detecting small kidney stones in ultrasound images presents additional challenges to detection of all sizes. The limited pixel representation and overlapping characteristics of larger stones make accurate identification difficult when attempting to find smaller stones. This leads to reduced sensitivity or failure of many detection models to identify the smaller stones which, if detected successfully, would also enhance the clinical utility and performance of the detection models.

A further considerable issue in detection models is the generation of false positive; no stone regions being misidentified as being affected by kidney stones and while improvements have been made to detection architecture, these models may still struggle to correctly identify stones due to the presence of noise or similar structures. This can potentially undermine the reliability of automated systems to provide accurate data and thus limit the utility of such systems in the clinical setting.

Achieving an accurate and efficient balance of accuracy and efficiency for detection of objects in a given scene continues to be difficult. The performance of contemporary architectures is based on large datasets created through a variety of data collection methods and conditions, which can affect their overall performance. Therefore, the variability in clarity and consistency of images within real-time ultrasound environments causes the performance of models to be dramatically hindered by these variations, indicating the necessity for more robust detection methods.

In this section, we will cover some of the restrictions imposed by the availability and quality of datasets when using ultrasound imaging to identify kidney stones. One major drawback that researchers face is their inability to access a large dataset that has been properly labelled to help build an accurate algorithm. Many studies currently depend on small datasets that have either been created or publicly shared, but these do not adequately represent the types of clinical scenarios encountered in practice.

Another major issue facing the point of view of having an imbalanced dataset is that typically “normal” images will occur in a greater volume than images related to the presence of kidney stones. This sort of imbalanced distribution can give rise to biasing a model toward making predictions of the majority class and consequently affect how well a model will be used for accurately identifying stones. Techniques like pseudo-labelling have been researched as part of a solution to this issue, however, their success still depends on having an initial set of high-quality labelled data for training the model.

Ultrasound images provide high levels of noise and variability that hinder the ability of models to appropriately identify useful features during training. Ensemble methods can add robustness to models, but they also introduce further computational complexity and do not fully eliminate the effect of having low-quality data on the model's prediction performance.

Also, the absence of standardized datasets across various research studies continues to be a significant limiting factor. Most datasets are generated using a variety of methods for the imaging modality (imaging techniques), equipment settings and characteristics of patients, making it difficult for models that have been trained on one dataset to provide reliable performance when tested against another, indicating the difficulties of obtaining reliable and consistent performance across multiple environments.

This domain lacks real clinical data that is publicly available. While publicly available datasets are helpful for first-time experiments, they don't typically contain as much diversity or complexity as clinical environments. Also, the raw ultrasound data will also need to be pre-processed carefully such as through selecting good high-quality frames prior to training a model.

The limitations of the dataset on which a model is trained have a significant contribution towards how well that model will perform and how well it generalizes. Limitations such as dataset size, imbalances between classes in the dataset, noise, and a lack of clinical data all pose challenges for future development of more accurate and robust kidney stone detection systems. These challenges lead to the need for improved data collection methods, more accurate annotation practices, and the inclusion of real clinical datasets in order to make kidney stone detection systems more reliable and robust.

An additional key challenge in data science is imbalanced datasets, where there are typically many more normal images than there are stone-positive examples. Such imbalances may result in bias towards 'normal' objects during model training and lower levels of performance when detecting objects from under-represented classes. As an example, Lin et al. [45] demonstrated how focal loss can address class imbalance during detection tasks but they also note that imbalanced data distributions are still a major obstacle in medical imaging technologies today.

A key problem faced is dataset imbalance, since there are far more normal images than there are stone-positive images in the dataset, such that the model could become biased towards predicting the majority class, ultimately leading to poor performance for the minority class. Despite many different approaches to deal with class imbalance, it continues to represent a major barrier for medical imaging applications.

Variability and noise will also pose serious challenges to the quality of ultrasound images. How well different ultrasound models are able to learn and extract useful features from these variables (using advanced feature extraction techniques) will be dependent on the quality and repeatability of the input data used during training of these models.

Moreover, another major limitation is that there are no adequate real-world, clinical datasets available to be used for both developing and validating. Publicly available datasets tend to be used, but don’t provide the necessary complexity or diversity that one would see in a real clinical context. This further emphasizes the need to improve upon the current data collection, annotation and integration processes to achieve better model generalization and practical applicability.

A further significant concern lies in the issue of domain shift due to variations between the different types of images captured, including but not limited to angle, quality, and method used for capture, can lead to degradation of model performance when the data distribution has changed from that upon which the model was trained. Thus there is a need for an improved approach to develop training techniques that allow for generalization to different sets of data.

This portion focuses on the shortcomings regarding transparency when using a deep learning model. Since most models tend to function like black box, it makes understanding how to interpret their output extremely challenging, and consequently decreases users' confidence in the output of particular applications that require the ability for human inspection.

While some techniques, such as Grad-CAM, create visual explanations, they do not always provide accurate visual representations for clinical decision making. For example, the heatmap generated by a Grad-CAM technical might include irrelevant areas instead of the correct area of interest. Hence, these limitations indicate that more reliable techniques of explanation are needed.

This part discusses the restrictions related to using deep learning systems in entire clinical systems. While there are many studies that have shown increases in the accuracy of detection, very few have focused on the actual application of these systems to day-to-day patient care. Consequently, most of these systems continue to exist at an experimental level without being used in actual healthcare systems.

One additional obstacle is the unavailability of user-friendly systems that can aggregate detection, visualization, and reporting into one unified pipeline system. While many models work well for testing purposes in laboratories, they are not created to function well in a clinical setting. In summary, this results in a disconnect between model creation design and the practical application of those models in a real-world clinical setting.

This study aims to overcome the limitations identified through the creation of a deep-learning-based system that will allow for accurate detection of kidney stones in ultrasound images. It specifically aims to improve detection accuracy, reduce the effects of variability in the database, and improve generalization through use of real clinical data. The addition of explainability methods will also enhance the reliability of the system in practice and support use of the new system in clinical applications.

Chapter-3 Methodology

This chapter provides the methodology used in this study which includes the proposed framework of the system, details of the dataset used, techniques for preprocessing the data, how to develop the models, and technology tools to be used. Additionally, an outline of how the kidney stone detection system is developed is provided as part of these sections of work.

3.1 Proposed System/Framework

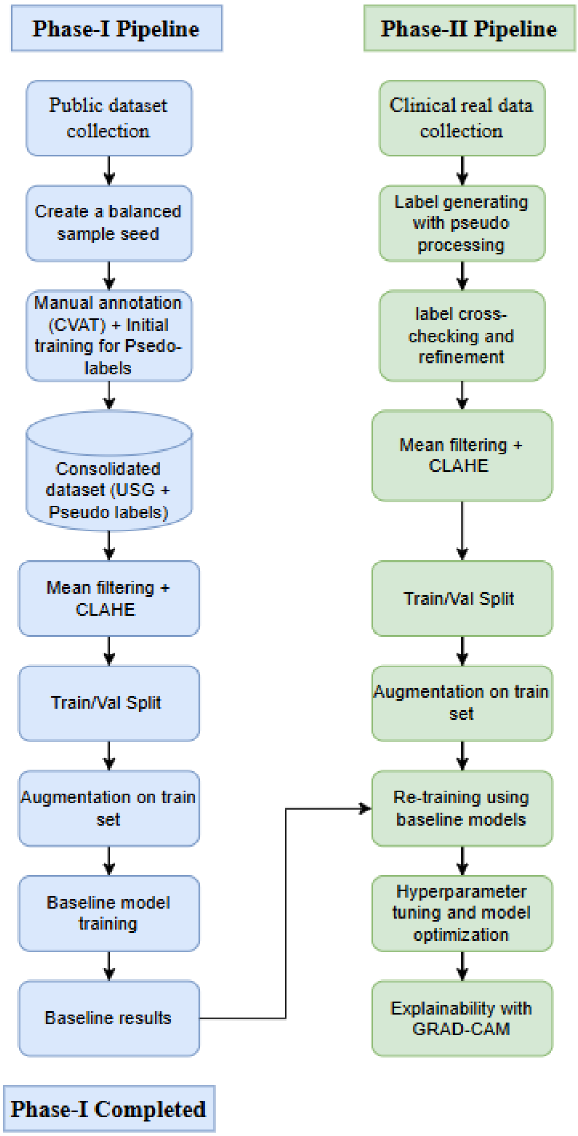


Figure 3.1 Overall framework of Phase-I and Phase-II pipeline

A deep-learning pipeline has been outlined to create a structured, reproducible system for renal stones detection via ultrasound imaging. The pipeline integrates the various steps required to perform these tasks, including data preprocessing, data annotation, pseudo-labelling (i.e., generating labels from existing datasets), model training (training models based on training datasets), and evaluation of models after training completion into one unified workflow.

There are two phases to the development of this framework; Phase 1 is the initial development and construction of the deep learning pipeline using publicly available ultrasound datasets and has been referred to as the "proof-of-concept phase". Phase 2 consists of the validation and expansion of the framework with actual patient clinical data (i.e., from clinical usage). The entirety of the framework is visually represented within Figure 3.1.

Multiple sources gather information into this system in order to allow for both growth and being able to provide accurate clinical data. Phase 1 has one publicly available database with 9,400 ultrasound images (some being positive for stones while others aren't), which provides a balanced database from which to generate an accurate machine-learning algorithm. The second phase uses collected ultrasound images through healthcare providers, which provides an opportunity for real-world data collection under conditions that include real-life noise, variability, and inconsistencies not seen in curated databases.

The seed dataset is created after data has been collected. For purposes of using the seed dataset to start training the model there is no need to manually annotate the entire dataset but to create a smaller and balanced subset of images that can be used as the initial labelled dataset. The seed dataset consists of both stone positive and normal cases across differing imaging conditions. The rationale for this step is to minimize the effort required to manually annotate as well as to ensure that there is an adequate and diverse representation of all types of samples to use for training the initial model.

Once annotation was completed, a baseline YOLOv8 object detection model was developed using the manually annotated seed dataset. The baseline model is the basis of continued data expansion, which was accomplished by using pseudo-labelling. To generate bounding box predictions for the remaining (unlabelled) images, the trained

YOLOv8 model is utilised. The predicted labels (pseudo-labels) are combined with the manually labelled data to greatly increase the amount of labelled data available.

Combining both manual and pseudo-labelling creates a single data source. This is used as one of the main sources for future modelling. To guarantee trustworthy data, the pseudo-labels undergo very basic quality checks to eliminate any incorrect or inconsistent annotations. This also aids in preserving the total quality of the dataset, while also taking advantage of the scalability provided by the use of pseudo-labelling.

Once the merged data file has been generated, a standardized approach to preparing data was conducted so that all data would have the same level of consistency during the two phases of the project. The merged data file was split into two sets – a training set and a validation set – using an 80/20 split. A fixed random number seed was used to guarantee that the results of both the experiments and subsequent analysis can be reproduced. This phase allows the model to be developed using a sample representative of the entire dataset, and at the same time allows for an unbiased evaluation of performance on the validation dataset.

Data augmentation approaches can enhance the robustness of the model and its ability to generalize among different ultrasound conditions by introducing variations into the training data. Implemented as part of the data augmentation methods, the training data is augmented through various operations (horizontal flipping, rotation, scaling, brightness adjustment, etc), and therefore will provide different examples of input for training purposes to allow for the model to learn invariant features as well as to generalise effectively under various imaging, contrast, noise, and orientation conditions using ultrasound.

After the dataset has been fully developed, model training was initiated on the consolidated labelled dataset. Phase one of model training created baseline models using both the manually annotated and pseudo labelled public datasets combined into a single dataset. A variety of deep learning models was developed, such as convolutional neural networks for classification and object detection models, to provide initial performance benchmarks of the models. Baseline models then served as the starting point for further improvements made during phase two.

Phase-I outputs, such as trained model weights and detected results, are used in Phase-II. Rather than re-train the models from ground zero (Phase-I), Phase-II was used to further adjust and enhance the previously trained models using real-life clinically based imaging (e.g., real images). In doing so, the system's ability to adjust to greater complexity and noise in imaging was used with previously trained characteristics to enhance existing trained systems.

Phase II involves processing clinical ultrasound images through a unified pipeline for consistency. Pre-trained models were then assessed for their performance on clinical data; this is the stage where predicted labels would undergo careful reviewing to ensure quality of the annotated labels and possible improvements through refinements were made to enhance both dataset performance as well as model performance.

Additional optimization to the original models with clinical data can be accomplished during this fine-tuning process. Specifically, optimizing hyperparameters (e.g., the learning rate, the batch size, and the number of training epochs) in order to improve model detection performance and reduce overfitting is critical at this stage to enable the models to generalize to an actual clinical environment where the ultrasound images may vary in quality.

To make clear what the model predicted, explainability techniques were used. Grad-CAM was used to produce heatmaps that showed which areas of the image are involved in making the prediction. This allowed us to view visually if the model is attending to clinically significant areas, and thus increase our confidence and reliability in the system.

3.2 Dataset Description

The accuracy of imaging-based medical deep learning model-based prediction is directly related to how good, diverse, and representative of an actual population is the data set used to train and validate the deep learning model. In this study, to make sure that the proposed system is scalable and applicable in using images from both datasets, one is publicly available and used in phase 1 to help develop the initial application — an image data set of ultrasound images, the second is a clinically acquired dataset, used in Phase II to validate the application experimentally and for actual online deployments following expected outcomes.

The initial development and training of novel ultrasound-based models was performed using a publicly available dataset containing approximately 9,400 ultrasound images with both stone-positive and normal cases. The images were collected under controlled conditions and provided a balanced distribution of classes, which is critical for training robust machine learning models. These images were retrieved from an open-source repository (available through a web browser at: https://data.mendeley.com/datasets/h6jc4xm4py/1) and were used extensively throughout model-building activities, including baseline model development and implementing pseudo-labelling techniques to augment the dataset. Sample images from the dataset are shown in Figure 3.1.

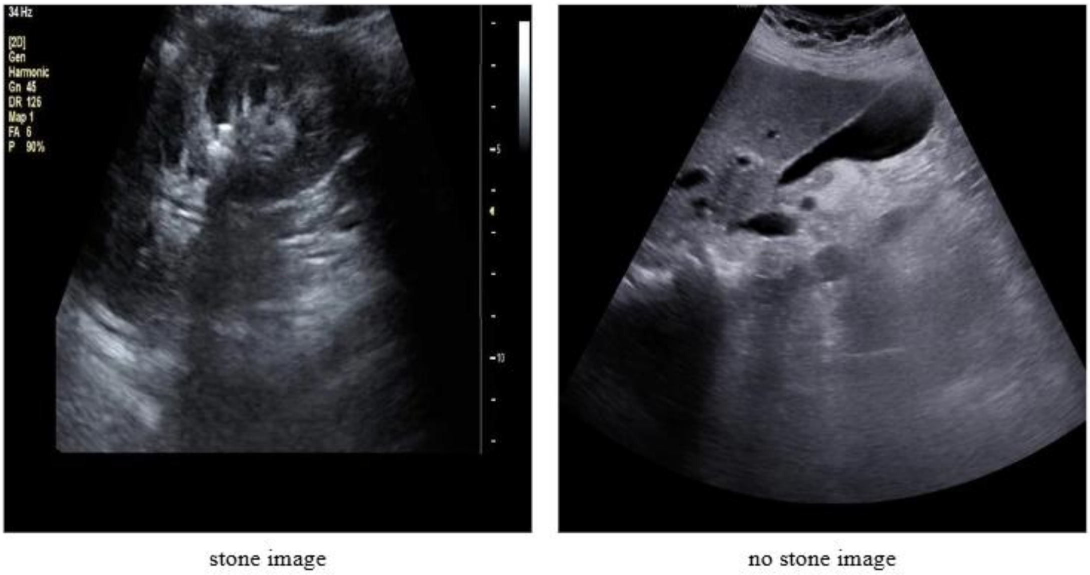
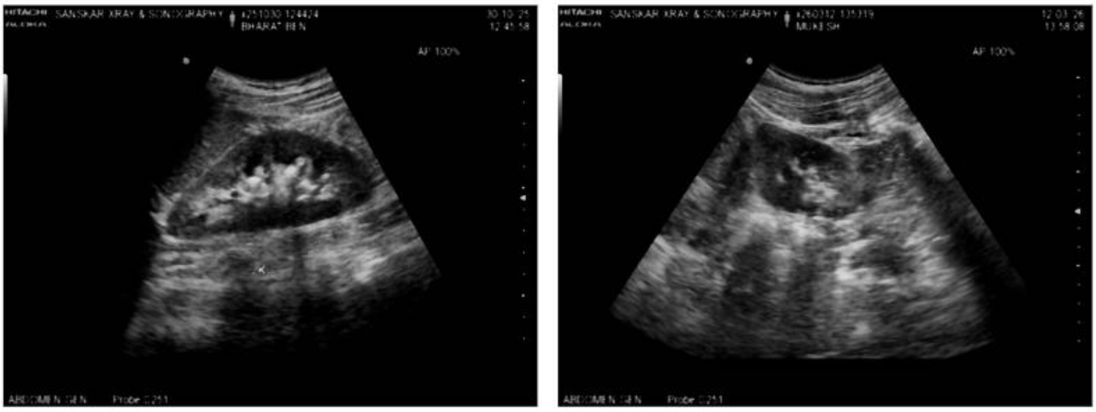


Figure 3.2 Sample Images from Public Dataset Showing Stone and No Stone Cases

An additional clinical dataset was collected in Phase II of the study to help assess how well the new system will work when used in practice. This dataset was obtained at the Sanskar X-ray and Ultrasound Clinic located in Patan, and Dr. Manish Patel, a radiologist for the lab, assisted in collecting the data. There are a total of 1726 ultrasound photos, with 796 showing renal stones and 930 without stones. Sample images are shown in Figure 3.2.

The clinical dataset had the added difficulties imposed by using photographs taken under real-life situations compared to the publicly available datasets. These challenges included speckled noise, different amounts of contrast in the photographs, and different techniques for obtaining those photographs. Because of these factors, the clinical dataset should much more accurately reflect actual practice and will be of great assistance in properly evaluating the robustness and reliability of the models that have been developed.



stone image no stone image

Figure 3.3 Sample Images from Real Dataset Showing Stone and No Stone Cases

To organize and structure the features needed for classifying and detecting images of renal stones, we developed an approach for both classification and detection. Classification involved dividing images into two classes (stone and no stone). For detection, we created bounding box annotations for locating renal stones in the images. Images with no renal stones had empty label files so that they could work with the detection framework.

Both public and clinical datasets were utilized to fully evaluate our proposed system. By using a public dataset for model development and controlled experimentation, we were also able to use the clinical dataset to validate results under realistic conditions. The dual-dataset approach allowed us to produce accurate models and generalize across diverse distributions of data.

3.3 Data Preprocessing

The preprocessing of data is important to have better-quality ultrasound pictures and also help the efficiency of deep learning models. Because ultrasound image characteristics are impacted by speckle noise, poor contrast, and artifact, they often contain many details about structures like kidney stones that are difficult to see in the presence of these types of noise. Therefore, all images underwent a systematic pipeline for preprocessing to enhance their clarity, consistency, and fitness for training a model.

The initial phase for processing the ultrasound images was to convert them from colour to grayscale. As ultrasound images mainly include intensity-based data, instead of colour based data, converting them to grayscale reduced how complex and/or how much processing is required by reducing the amount of information that must be processed while still ensuring that all of the subsequent processing performed is based on structural as well as intensity-based information, both of which are required for renal stone detection.

Median filtering was used to decrease noise after converting images to grayscale. This technique is particularly well suited for removing "speckle noise" from ultrasound images while preserving the fine details of edges. Unlike linear filters, median filters replace every pixel with the median of that pixel's neighbours, which effectively removes the noise in the image without significantly blurring important features.

$$I'(x,y) = median\{\, I(i,j) \;\mid\; (i,j) \in N(x,y)\,\} \tag{1}$$

The surrounding pixels of pixel (x, y) are denoted as (i, j) where i is an index into the neighbouring pixel data array around pixel $(x,\ y)$ and j is used to identify the neighbouring pixel itself. The purpose of the spatial filtering is to reduce the amount of speckle noise that may be present in ultrasound images while retaining sharpness of any edge areas for the purposes of enhancing the detection of renal stones [11].

CLAHE (Contrast Limited Adaptive Histogram Equalization) was used for contrast enhancement after noise reduction. This technique applies histogram equalization to small areas instead of the whole image, so therefore improves local contrast within specific areas. Contrast limiting also limits noise amplification.

$$CLAHE = \left(\frac{(C-Cmin)}{Cmax-Cmin}\right) \times (L-1) \tag{2}$$

Contrast limited adaptive histogram equalisation achieves an improvement in local contrast by using histogram equalisation over the small area of the original image, and then limiting the amount that the enhancement should occur in order to prevent from over-enhancing the existing noise [5],[11]. The effect of preprocessing on clinical ultrasound images is illustrated in Figure 3.2 and 3.3 for phase-I and phase-II.

Apart from enhancing visual quality, preprocessing is essential for improving the capabilities of deep learning algorithms when used to analyze ultrasound images. By using median filters, speckle noise is reduced; speckle noise is a common artifact found in ultrasound images that can hide small details such as small stones in the kidney. With the application of the contrast-limited adaptive histogram equalization (CLAHE), local contrast is enhanced enabling subtle intensity differences in the image to become more easily detected. Therefore, improvements in both contrast and noise will allow the underlying structures in the ultrasound image to be seen more clearly.

Preprocessing enhances the visibility of renal stones that show up in X-ray images as small, bright spots. This allows for deep learning models to learn better features and classify stone and non-stone areas more accurately. Preprocessing also provides consistency among different image sources (between clinical and public datasets) that helps minimize bias in model training and to increase generalization across different types of image acquisition conditions.

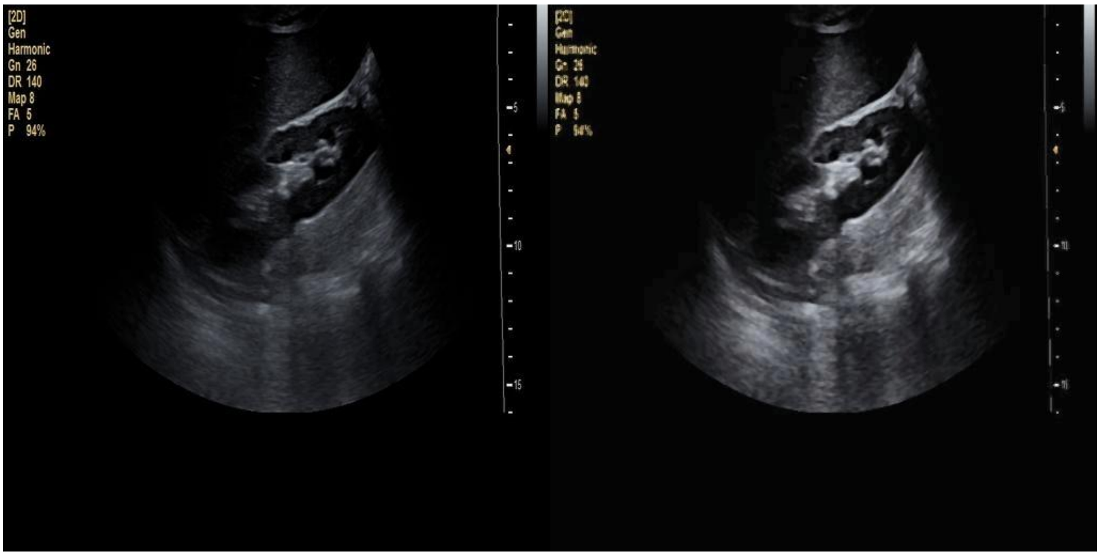


Figure 3.4 The Result of Image Pre-Processing Techniques Used in Ultrasound Imaging: Phase I Dataset

To improve the resilience and generalization power of our deep learning model, we also applied data augmentation techniques in addition to noise reduction and image contrast enhancement. Imaging equipment characteristics, patient anatomy, acquisition angles, and operator settings can all greatly affect ultrasound image acquisition, therefore the images in our training dataset needed to have an even greater degree of variability to

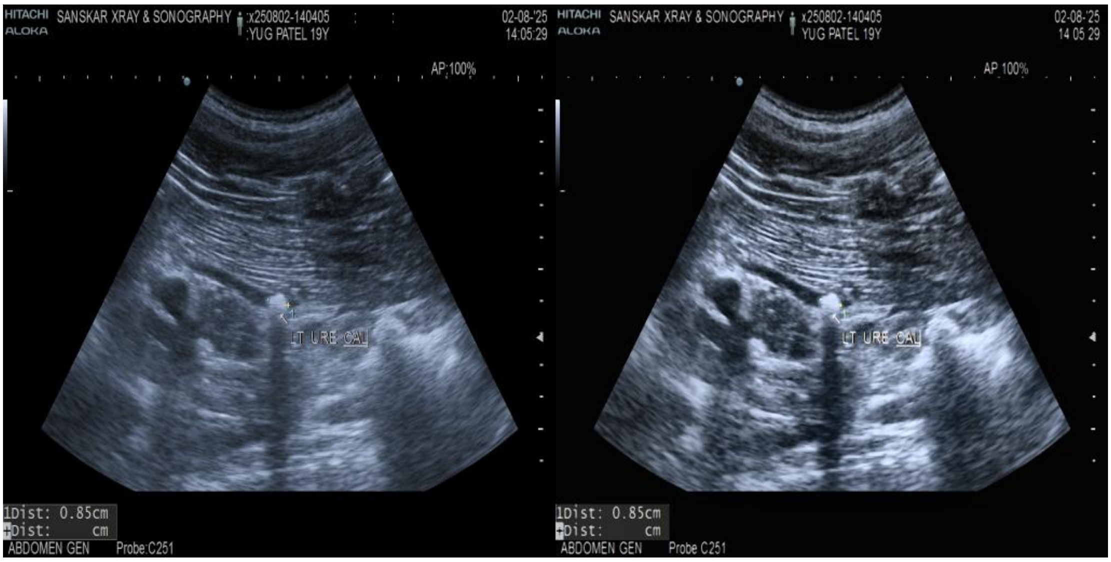


Figure 3.5 The Result of Image Pre-Processing Techniques Used in Ultrasound Imaging - Phase II Clinical Dataset

counteract any potential issues with overfitting that may arise from the variances introduced by imaging. Data augmentation was performed only on the training dataset to address this issue.

Augmentation of the input images consisted of applying many geometric and intensity-based transformations. The geometric transformations were horizontal flipping, rotation, and scaling. The intensity transformations were adjusting the brightness of, or changing the brightness in, the image. Horizontal flipping allows the model to learn variation based on symmetry, while the rotation and scaling transformations mimic variations in the orientation of the probe or the image acquisition of the probe. Brightness adjustments also assist the model in adapting to differing illumination and contrast levels found in ultrasound imaging frequently.

In addition to the above-mentioned changes, small variations in noise were also added during the training process to mimic real-world ultrasound conditions. This step was crucial for increasing the ability of the model to perform consistently on clinical data; since there are more artifacts and inconsistencies in the imaging of clinical cases than in publicly available data sets.

Exposing a model through augmented images will develop a more robust training process by allowing it to create features that can remain invariant to the different orientations, scales and intensities. This will result in improved generalization to examples of previously unseen data, thus improving the performance potential of a model in real-life clinical situations.

It's crucial to mention that only the training data received augmentation of any sort. The validation data set, however, did not receive augmentation at all. Because the model is evaluated on original and unaltered data, there are no biases inherent in the performance evaluation.

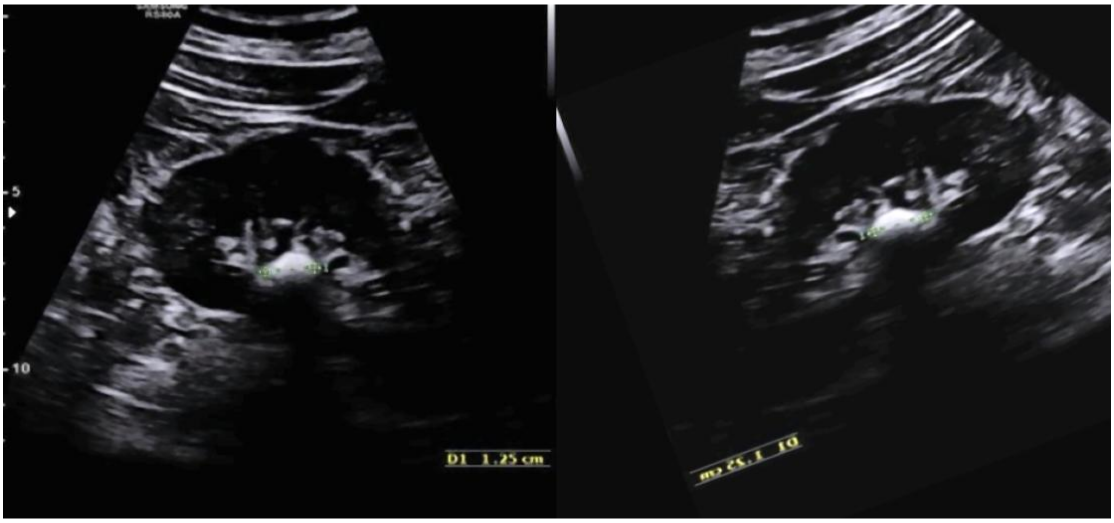


Figure 3.6 Augmented ultrasound image sample – Phase-I

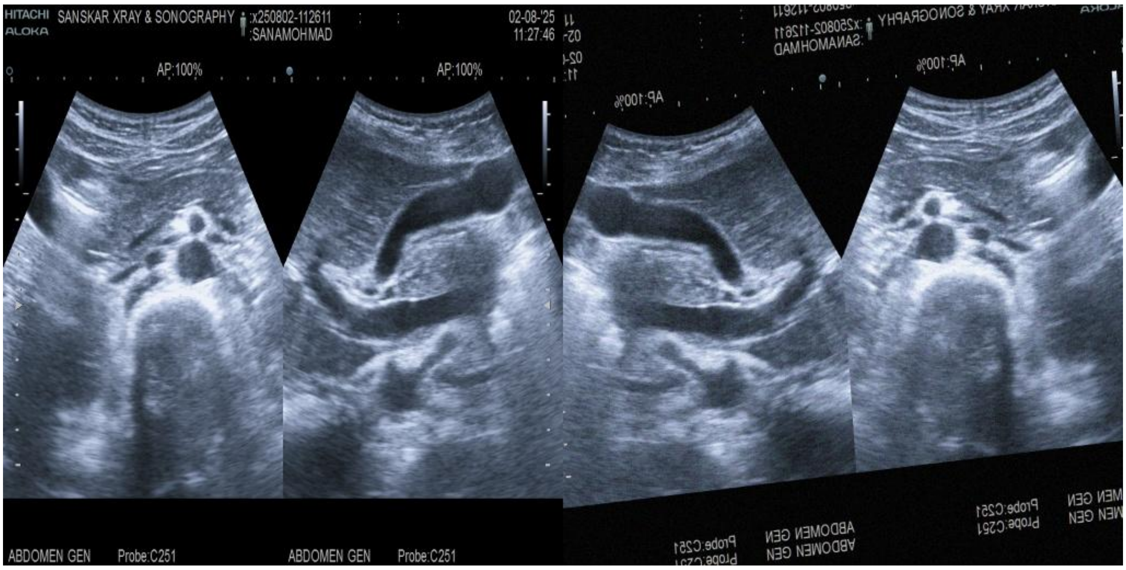


Figure 3.7 Augmented ultrasound image sample – Phase-II

3.4 Feature Engineering / Selection

Feature engineering is an essential component of Standard ML Coding as it helps us select and derive the relevant features from the input data. This study was conducted at a time when traditional and automated methods for determining the appropriate features are both possible with the use of deep learning algorithms and previously, there were no manually defined or hand-crafted features for use within the deep learning models. The additional use of ultrasound images (which typically do not lend themselves to manually defining features as a result of poor-quality images, low echo contrast, etc.) offers opportunities for an additional benefit as well.

Traditional approaches to machine learning rely on human-selection of various features including texture, intensity, and shape descriptors to provide input to classification algorithms; which can work well if conditions are controlled but tend not to generalise as well with more heterogeneous datasets. Ultrasound images pose an even greater challenge; they have lots of problems due to having superior amounts of speckle noise creating uncertainty regarding boundary definition related to all manually created feature extraction methods.

Unlike other types of models (e.g. traditional machine learning), deep learning methods learn to construct hierarchically structured representations for input data automatically. A CNN model learns low level characteristics of an image (edged/textures) and combines those learned characteristics to identify higher level characteristics that correspond to complex features associated with kidney stones from images of those stones. Automated learning of features allows deep learning models to eliminate the need to rely on manually designed (or otherwise pre-defined) feature sets for each unique type of kidney stone in the dataset, and will also provide improved model accuracy by accommodating for more variability within the dataset.

Structured annotations improve how an object detection task is represented as a feature. The dataset was prepared in a particular format using YOLO for object detection with annotations applied to each image with bounding boxes that indicate the location of kidney stones. Each annotation comprised a set of normalized coordinates representing the centre position, width and height of the bounding box and the class label. When no stones were located in an image, the empty labels (no bounding box), were used to ensure compatibility with detection frameworks.

Moreover, bounding box labels were also verified and revised within the same label set for accuracy and consistency. This was done to ensure that training data accuracy and consistency were maximized as inaccurate training data may adversely affect the learning and performance of the model being trained in terms of both classification and detection. The hybridization of automated feature learning with structured labelling guarantees that all features given to the models as input to perform either classification or detection will be of value and have reliability associated with them.

3.5 Model Development

Model development includes the design, training, and evaluation of Deep Learning models for the classification/detection of renal stones on ultrasound scan images. The proposed system combines Convolutional Neural Networks (CNNs) with Object Detection models to identify and localize renal stones. The model development process took place in two phases of development. Phase-I involved training models on publicly available datasets, while in Phase-II, these models were refined and validated using clinical ultrasound data.

During Phase I, baseline performance was established through studies of multiple deep learning architectures. For classification tasks, different types of convolutional neural networks were tested; and for localization tasks, that included object detection models. Classification models would determine if there were renal stones in the ultrasound images, and the models used to detect spatial locations of stones were also created from the ultrasound images.

3.5.1 Classification Models

Ultrasound images can be classified as "stone" or "no-stone" using classification models. To develop these classification models, two main architectures were used: ResNet-18 (CNN2D) and EfficientNet-B0. To learn domain-specific features related to ultrasound images directly from the data, both models were trained from scratch.

A. CNN2D – ResNet-18

ResNet-18 is a type of Convolutional Neural Network used as a baseline for classification. Residual connections are included in ResNet, which enables the neural network to learn identity mappings in addition to the usual classification output of traditional CNNs. The inclusion of residual connections alleviates the vanishing gradient issue and makes it possible to effectively train much deeper architectures than previously possible. [35]

$$y = F(x, \{W_i\}) + x \tag{3}$$

Here, the input feature map is referred to as x and F(x) represents the residual mapping.

In phase I the ResNet-18 model was built from nothing as opposed to starting with pre-trained weights. Only the final fully connected layer was changed to have two outputs (stone/no stone). During this training phase, we used a dropout of 0.7, which helped reduce overfitting and enhance overall performance. For the training of phase one we used the Adam optimizer (lr = 0.001) with a batch size of 32 for a total of 15 epochs. In addition to the training data, we used random cropping, flipping and brightness adjustments to help create stronger and more robust models.

In phase II, we re-trained the model with the same architecture using a clinical dataset in order to test the results of using ultrasound images on a larger scale. We used a higher dropout (0.8) for this second phase of training as a result of increased variability and noise associated with typical ultrasound images. We also continued with the same training parameters (i.e., 15 epochs) as in phase one but selected the optimal weights for the best performing model based on validation accuracy.

B. EfficientNet-B0

Because of EfficientNet-B0's ability to achieve high accuracy with fewer parameters (through 'compound scaling') in terms of depth, width and resolution, it was chosen to represent an advanced classification model. Its unique ability (i.e., computational efficiency and accuracy) makes it especially suitable for use in medical imaging tasks. EfficientNet's compound scaling approach uniformly scales a model's depth, width and input resolution using a set of fixed-scaling coefficients [36]. The following equation illustrates this formulaic relationship:

$$D = \alpha^{\varphi}, \quad w = \beta^{\varphi}, \quad r = \gamma^{\varphi}$$

$$\text{subject to} \quad \alpha \cdot \beta^2 \cdot \gamma^2 \approx 2 \tag{4}$$

where D, w, and r denote the depth, the width, and the network's resolution respectively; and ϕ will be the scaling factor that is chosen by the user. The formulation is designed to provide balanced scaling and maintain computational efficiency.

In the case of this research, the EfficientNet-B0 model was used as-is with no change in the internal architecture. During Phase I of the study, the model was trained completely from zero with modifications made to the classification layer so that it produced two classes (i.e. renal calculi/not renal calculi). In addition, a dropout (0.6) layer was added to reduce overfitting. The model was trained using the Adam Optimizer and the parameters for the training included number of epochs (30), batch size (32), and learning rate (0.001). In addition, a learning rate schedule, ReduceLROnPlateau, was employed to dynamically adjust the learning rate based on validation loss.

For Phase II of the study, the model was retrained on the clinical dataset using the number of epochs (15) and dropout (0.5) which had been adjusted to provide a balance between model complexity and generalization performance in real-world settings. The same optimization strategy was used as was used in Phase I.

Using the EfficientNet-B0 and ResNet-18 (conventional architecture vs computationally optimized) models allows for a better comparison between the two methods of classifying ultrasound renal stones, therefore gaining a clearer picture of classification performance across these two methods.

3.5.2 Detection Models

Object detection models are needed to localize the location of stones within ultrasound images, while a classification model will identify those stones as present in the images. In this study, we used both the YOLOv8 (You Only Look Once) object detection model and the RT-DETR (Real-Time Multi-Object Detection Model) object detection model because of their ability to accurately and efficiently detect objects in complex MRI data.

A. YOLOv8 (m and l)

YOLOv8 is a one-stage object detection system that does classification and localization in one forward pass, which allows for extremely fast real-time applications [37]. It works by directly predicting bounding box positions and class probabilities from the input image, i.e., it requires no region proposal step.

$$\mathrm{L} = \mathrm{L}_{\mathrm{cls}} + \mathrm{L}_{\mathrm{box}} + \mathrm{L}_{\mathrm{conf}} \tag{5}$$

Where L_{box} represents bounding box regression loss, L_{cls} represents classification loss and L_{conf} represents objectness loss. For Phase I, the developed YOLOv8 models (medium YOLOv8m and large YOLOv8l) started with pre-trained weights and were trained using the training dataset (prepared dataset). This phase consisted of 50 epochs with each model receiving image inputs of 640 × 640 pixels. The YOLOv8m model batch size was set to 16 while the YOLOv8l model used a smaller batch size of 8,

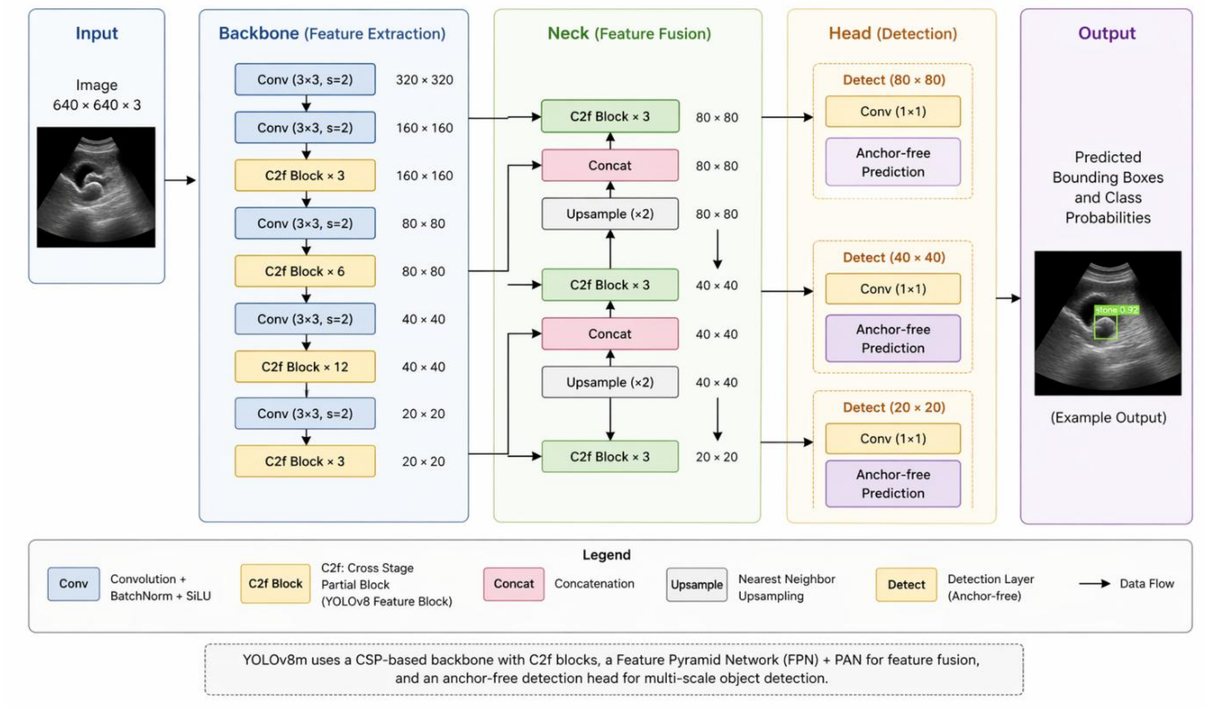


Figure 3.8 The Architecture of the YOLOv8 Model for Object Recognition

resulting from increased computation requirements for the YOLOv8l model. In Phase II, both YOLOv8 models were fine-tuned using the clinical data to enhance their performance in the real-world clinical application. Finished training weights from Phase I of each model were used to initialize each respective model and were set to an updated learning rate of 0.0001 (YOLOv8m) to allow both models to maintain retained prediction capabilities from Phase I while accommodating the clinical data. Early stopping with a patience of 15 epochs was utilized as a stop condition in both models to reduce the possibility of overfitting.

during phase-II, augmentation techniques included change in brightness, rotation, horizontal flip, and mosaic augmentations to replicate different ultrasound conditions. similarly, the YOLOv8l model was finely tuned with a batch of size 4 because of the increase size of architecture.

because of their accurate and efficient detection of renal stones, YOLOv8 models are also suitable for real-time use in a clinical setting.

B. RT-DETR (Real Time Detection Transformer)

The object detection model (RT-DETR) uses a transformer architecture and is based on using attention mechanisms to be used to obtain context information over the entire image from a distance. Therefore, it is able to model long-distance dependencies on the same object. This means RT-DETR will be more capable of performing word recognition in images than a convolutional model that is limited to short-range dependency models [9].

$$\mathrm{Attention}(\mathrm{Q}, \mathrm{K}, \mathrm{V}) = \mathrm{softmax}\left(\frac{\mathrm{Q} \times \mathrm{K}^{\mathrm{T}}}{\sqrt{\mathrm{d}}}{}_{\mathrm{k}}\right) \mathrm{x\ V} \tag{6}$$

In Phase 1, the RT-DETR-L model was pre-trained and initialized with pre-trained weights and was trained for 50 epochs with an image size of 640 × 640 and a batch size of 4. Deterministic training settings were set to increase the reproducibility of results.

In Phase 2, the model was fine-tuned using the clinical dataset but maintained the same training setup as in Phase 1. The training configuration used a batch size of 2, due to the increased complexity of using transformer-based architectures. The use of RT-DETR provided global features in improved global feature representation at the expense of using more resources than comparable YOLO-based models.

3.5.3 Explainability using Grad-CAM

In medical imaging, it is important to have an understanding of how the model formulates its predictions; thus, there is a need for explainability in this domain. In this study, we utilized Grad-CAM to visually display the portions of an ultrasound image which contributed to the model's decision. Grad-CAM produces a heatmap that indicates which areas of the input image are significant to the model by applying gradients of the output class back to their respective inputs via the last convolutional layers of the model. [38]

$$L^{c}_{\{Grad-CAM\}} = ReLU\left(\sum_{k} \quad \alpha^{c}_{k} A^{k}\right) \tag{7}$$

Where, A^{k} represents the feature maps and α^{c}_{k} represents the importance weights assigned to each respective classification or segmentation.

Phase I utilized Grad-CAM to visualize how the model was predicting images in the validation set to begin to gain insights into how the model behaves in relation to the data set.

In Phase II, a more refined application of Grad-CAM with respect to the predicted bounding boxes created much more localized heatmaps than those created in Phase I were generated. Additional processing methods were also applied, such as using region of interest weights and creating foreground masks, thus improving the clarity and interpretability of these visualizations.

As a result, output from this process was visible as three-panel visualizations consisting of the original image, the generated heatmap and the overlapped version of the original image/heatmap. Verifying these visualizations for clinical relevance of the models increases our confidence in the system.

3.6 Tools and Technologies Used

A variety of advanced tools, frameworks, and libraries were utilized to create and implement the proposed renal stone detection system, which allowed efficient processing of data and for the development, training, and visualization of models. Due to its ease of use, versatility, and expansive collection of machine learning and image processing libraries; Python was selected as the main programming language for the system's implementation. The ability of Python to provide abundant support for deep learning frameworks greatly enhanced the functionality of the complex models which were developed throughout the research process.

PyTorch was selected as the primary deep learning framework for use in developing the model because of its support for dynamic computation graphs and efficient GPU acceleration, both of which are essential for training convolutional neural networks (CNNs) and transformer-based architectures. As such, PyTorch was utilized to train all classification models throughout the project, including, but not limited to, ResNet-18 and EfficientNet-B0.

To create object detection models (including YOLOv8), the Ultralytics YOLO framework was used which provides a simplified, user-friendly interface for working with these models across training, validation, and inference. It also includes many features that aid in configuring, augmenting, and visualizing datasets. This greatly improved the ability to create detection models using ultrasound image datasets.

The RT-DETR model was used with Ultralytics for transformer-based detection models, allowing testing of attention-based detection methods while still allowing integration with the current training system. Torchvision is another library based on PyTorch that aids in working with image datasets, including applying transformations and creating or obtaining architecture information (pre-defined). It was used to assist with preprocessing and augmenting ultrasound images while creating a detection model.

OpenCV was used for image-processing tasks such as reading images, resizing images, and applying preprocessing techniques. It was also important in implementing methods for explainability such as visualizations and overlays via Grad-CAM.

The pytorch-grad-cam library was used to implement Grad-CAM in order to provide interpretability for the models used. The library allowed for generating heatmaps that highlight areas of ultrasound images that are significant, thus helping to improve understanding of what the models were predicting. NumPy was used to perform numerical computations and array-based operations, while Matplotlib was used to visualize results, including plotting heatmaps and displaying processed images.

GPU accelerator for training and experimentation provided quick training times whilst managing large amounts of data more effectively. Other utilities like tqdm were used within the application to facilitate viewing training progress and improve usability throughout model training.

In summary, the tools/technologies combined allowed for creating a robust/effective system that could support both the classification/object detection of renal stones, along with providing explainability features.

Chapter-4 Implementation and Results

This chapter discusses how well the system works according to its defined parameters. Outlined within this chapter are: experimental setups, training/testing details, performance evaluation metrics, results of both Phase I and II and comparative analysis of the developed models.

4.1 Experimental Setup

This research corroborates that it is possible to develop a system that can successfully identify renal stones in both controlled conditions and real-world settings. To achieve this, a structured pipeline was created with four modules (preprocessing, model inference, detection, and explainability). Consequently, the above experiments were carried out in two phases, to assess both model development and validate the performance of the developed model on various data distributions. The programming language Python was chosen because of its availability of many machine learning and deep learning libraries. The primary deep learning library used was PyTorch. This offered versatility when building and training classification models, such as ResNet-18 and EfficientNet B0. Additionally, for Object Detection tasks, the Ultralytics YOLO framework was implemented to implement and evaluate YOLOv8 models, and for the transformer-based Object Detection tasks RT-DETR were used.

GPU acceleration was used for enhanced computing efficiencies and reduced training times during the training and experimentation of the system. The execution was performed in a CUDA-enabled, parallelized batch processing environment, thus allowing for large sets of ultrasound data images to be processed in parallel as well. This allows for both training of the detection and classification models, and for either of them to be trained within an acceptable period of time.

There are two phases to the experimental design; Phase I involved the development and testing of models using a publicly available ultrasound image sequence dataset, which will establish baseline performance and validate the capability of different models to extract important characteristics pertaining to renal stone detection from an ultrasound imagery data sample. Phase II involved continued testing of the trained models and refinement using a set of clinical ultrasound images obtained from a real-world clinical setting. This phase will be necessary to evaluate the generalizability of the models and how these models might be applied in a clinical practice setting.

Using a splitting ratio of 80:20, the dataset utilized in both phases was separated into training/validation sets. This allowed the model to be trained with adequate data while also allowing for an unbiased evaluation via a separate dataset. To maintain consistency and comparability of results, the same data splitting technique was applied throughout the entire experiment.

In order to compare the models fairly, an identical experimental pipeline was used. All input ultrasound images were pre-processed using the same techniques prior to being input into the models. As a result, in this manner, performance differences between models could be attributed to architectural differences and not to the quality of input data.

Figure 4.1 shows the total work processes of this system. The work of the system starts with the receipt of ultrasound images and then there are several preprocessing operations to enhance quality. After the images are enhanced, they are transmitted through models that classify and detect the existence and location of kidney stones. After the model detects the stones, Grad-CAM is used to analyze and generate visual explanations for model predictions. Last, the results created by using Grad-CAM will be displayed using an online browser (or web) interface that allows the user to access detection output and create structured reports from that output.

Both classification and detection models used in the same experimental framework facilitate a thorough evaluation of model performance together. The classification model determines whether or not there are kidney stones, and the detection model enhances the clinical utility of the classification model by determining the location of the stones on the ultrasound images. With both model types available to the clinician, they can assess diagnostic accuracy and implement practical applications for every individual.

The experimental design provides an additional feature of visual representation of the experiments completed; by using a web-based demonstration system. The web interface provides the user with the ability to upload ultrasound images, run detection algorithms on those images, view the Grad-CAM visualizations associated with those detections, and download reports of generated results. The incorporation of these features into the experimental design supports the practicality of the deployment of the proposed system and further illustrates the suitability of the system for use in a real-world application.

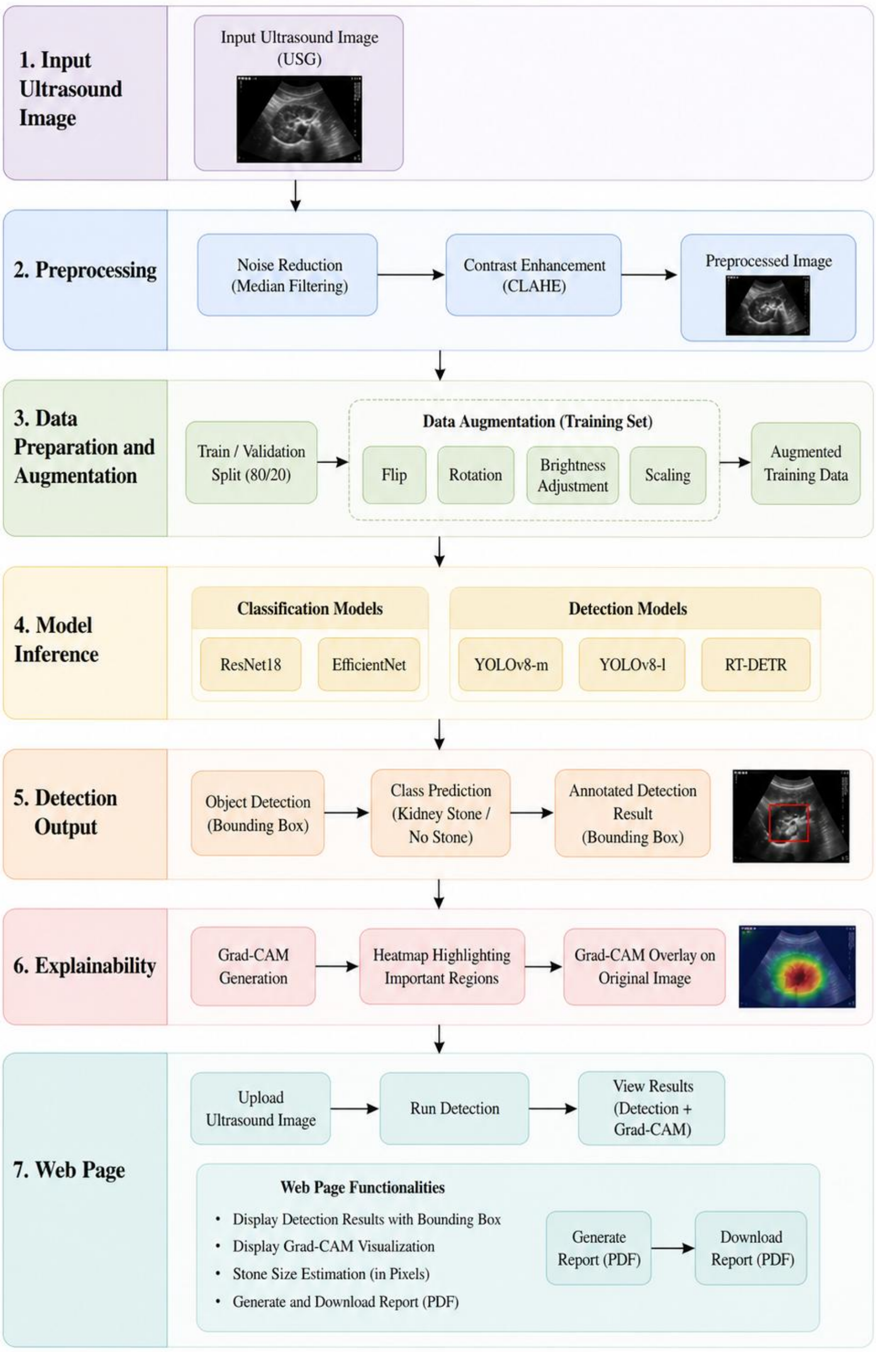


Figure 4.1 Overall Experimental Workflow

The experimental design was crafted with care so that repeatable experiments could be performed as well as offering a fair comparison between multiple models and relevance for actual use cases in clinical practice. This is accomplished through the use of structured experimentation in the laboratory setting and validation of the results in a clinical setting. With this type of experiment, there is ample evidence to support the development and testing of the proposed renal stone detection device.

4.2 Training and Testing Details

The evaluation of how well the new renal stone detection applications work, both in training and testing, was put through the same methodology and structure in order to assure comparable and reliable results across different types of models and all datasets used for the evaluation of these models. These evaluations were done for classification versus object detection applications separately, however the same conditions (both for pre-processing and evaluation) were followed.

4.2.1 Data Preparation for Training

In this study, the datasets were separated into Training and Validation datasets for both phases (Phase-I and Phase-II) using an 80:20 ratio (80% of the data used to train and 20% to validate). The splitting of the datasets enabled enough training sample data, while keeping separate datasets to evaluate the models' performance.

The dataset splitting was performed without any data leakage by ensuring that no images were overlapped between the training dataset and the validation dataset during the splitting process. All models and experiments followed the same splitting procedure to ensure that a fair comparison of the models' performances.

To enhance the robustness of the model and its generalizability, data augmentation approaches were used during the training phase. A complete description of the augmentation strategies used for this study are described in Section 3.4.

4.2.2 Training Setup for Classification Models

ResNet-18 and EfficientNet-B0 were the two models that we used to train the data to classify objects/categorize them according to characteristics found in ultrasound images; the training of these models was done with no previously trained weights in order to allow the models to determine what characteristics are unique to ultrasound images.

All of the input images were resized to a standard (fixed) resolution of 224 by 224 pixels so that they could be input into all model types. The training of the models was done with the Adam optimizer (initial learning rate of .001) and a batch size of 32 to maximize computing power that would be dedicated to training without affecting the stability of the training process.

The ResNet-18 model used dropout as a form of regularisation and was set at 0.7 in Phase I and 0.8 for Phase II to enhance its capability for generalising to new cases, particularly where there may be noise within the clinical setting. Likewise, the EfficientNet-B0 model implemented dropout with values of 0.6 in Phase I and 0.5 in Phase II. The rationale for altering the values was that training would produce an appropriate balance between the model's complexity and possible overfitting.

The number of epochs per phase is discontinuous. In Phase I, there were numerous epochs used for training the models to ensure sufficient learning from the public dataset while in Phase II, due to a small dataset with more variability in the value of each sample, training was limited to 15 epochs. Further to this, the EfficientNet-B0 model employed a learning rate schedule to modify the learning rate as determined by the models' performance on the validation dataset, allowing the models to converge with greater stability throughout training.

4.2.3 Training Setup for Detection Models

Using object detection models, kidney stones in ultrasound images could be detected both through identification and localization. YOLOv8-m served as the primary detection model used in this paper, with further comparisons between the performance of YOLOv8-l and RT-DETR being conducted as part of this study.

All input images were resized to 640 × 640 pixels in order to train the models, and each model was trained over 50 epochs to learn both the spatial characteristics of the ultrasound images. The batch sizes used during the training of the models, as indicated by their complexity, were also varied; YOLOv8-m had batch sizes of 16 and 8, YOLOv8-l had a batch size of 4, and RT-DETR had a batch size of 2 due to its increased computational needs.

The first phase consisted of using a pre-trained model on a general, publicly available dataset. The second phase consisted of fine-tuning this model using a clinical dataset to refine the weights from the first phase that yielded the best results. This transfer of features from one dataset to another allowed the models to converge quicker and perform better in real-world situations.

Fine-tuning during phase two used a lower learning rate of 0.0001 to keep previously learned features while allowing for adaptation to new data. Early stopping with a patience value of 15 epochs was also utilized in order to minimize overfitting and maximize the performance of the model.

4.2.4 Testing Procedure

Evaluation of trained models took place on the validation set to verify performance. The models were evaluated without applying augmentation techniques; thus, data evaluation was performed using original images.

All input images had the same preprocessing done as were done in training to ensure consistency. The output predictions from classification models were compared to the ground truth labels to assess accuracy, while detection models were evaluated by assessing their predicted bounding boxes, confidence scores, and how they align with the ground truth annotations.

The standardized testing methodology made certain that all models were evaluated under standard and controlled conditions when determining their effectiveness.

4.2.5 Implementation Consistency

The same implementation strategy for each model was employed to achieve an unbiased comparison across all models being tested. Every model was also developed from an identical pre-process pipeline, dataset separation strategy, and evaluation scenario to maintain the same conditions.

When conducting experiments with consistent implementations, any discrepancies in performance between multiple models can be attributed solely to the models' architectures and learning capabilities as opposed to differences among the experimental setups. Therefore, this level of uniformity across the experiments improves the credibility of the results obtained, resulting in more effective comparative analyses of the different models.

The consistency between models ensures that any difference seen in performance is solely as a result of model performance and is not a result of different experimental conditions.

4.3 Performance Metrics

We used standard metric evaluation methods to assess the effectiveness of our proposed renal stone detection system through both classification and object detection. The quantitative metrics provide a complete picture of the accuracy of the model and how well it performs in detecting stones and other objects.

4.3.1 Classification Metrics

To evaluate classification models, we used accuracy, precision, and recall as measures of how accurately predictions are made and how well a model can differentiate between images containing stones vs. images that do not. The overall accuracy is determined by calculating the ratio of correctly classified samples to the total number of samples in a classification model.

$$Accuracy = \frac{TP+TN}{TP+FP+TN+FN} \tag{8}$$

Precision is used to assess the accuracy of positive predictions by determining the proportion of positive predictions made correctly.

$$Precision = \frac{TP}{TP+FP} \tag{9}$$

Recall indicates how well the model has identified all actual positive cases, which is critical for medical diagnosis were failing to identify a stone may have negative clinical outcomes.

$$Recall = \frac{TP}{TP+FN} \tag{10}$$

4.3.2 Detection Metrics

Regarding the evaluation of object detection models, the following metrics were used to assess the accuracy of the predicted bounding boxes regarding their ground truth annotation: precision, recall, and mean Average Precision. Precision and recall for detection consider the accuracy of the predicted bounding boxes in terms of both the accuracy of the localization of the box and the accuracy of the classification (correctness of the predicted object).

The primary evaluation metric for measuring how accurately the predicted bounding boxes correspond to ground truth annotations is the mean Average Precision; this metric assesses the correctness of predicted bounding boxes at a given Intersection over Union (IoU) threshold.

$$mAP = \frac{1}{N}\sum_{i-1}^{n} = AP_i \tag{11}$$

In this research, mAP@0.5 will be utilized to assess the performance of individual detections. A detection is said to be accurate if the predicted bounding box intersects with its corresponding ground truth bounding box at least 50%. The 50% overlapping bounding box criterion combines both accuracy of the established localizations and practical detection performance.

4.4 Results

This section provides results of the renal stone detection system that have been validated with results from Phase-I and Phase-II experiments. In this section the classification and object detection models have been evaluated with visual evaluation through detection outputs and Grad-CAM visualizations. Results detailing the models' performance on publicly available datasets and clinical ultrasound images (real) have been included.

4.4.1 Quantitative Results

Standard metrics referenced within Section 4.3 were utilized to evaluate how well each model performed quantitatively. The quantitative performance results from Phases I and II experiments have been summarized within Table 4.1 & Table 4.2.

Phase I results indicate that both the classification and detection models have learned valuable features present within the ultrasound dataset. It is important to note that EfficientNet-B0 performs better than ResNet-18 in terms of accuracy, demonstrating EfficentNet-B0's effectiveness as a feature extractor when compared to other classifier's tested during this research.

Based on the performance results of detection models it appears that YOLOv8-m performed well across various metrics such as precision, recall, and mAP@0.5. As such it would be capable of being used for real-time detection tasks. RT-DETR also performed well, but this model requires significantly more computational resources to run than YOLOv8-m.

Table 4.1: Performance of evaluated models in Phase-I

Model	Precision	Recall	Acc./ mAP@50
CNN2D	0.969	0.965	0.9687
EfficientNet-B0	0.997	0.997	0.9969
YOLOv8m	0.822	0.726	0.8212
YOLOv8l	0.788	0.735	0.8094
RT-DETR-L	0.749	0.649	0.7357

Table 4.2: Performance of evaluated models in Phase-II

Model	Precision	Recall	Acc./ mAP@50
CNN2D	0.782	0.814	0.8035
EfficientNet-B0	0.931	0.942	0.9450
YOLOv8m	0.925	0.955	0.9798
YOLOv8l	0.936	0.960	0.9755
RT-DETR-L	0.899	0.921	0.9505

Phase II primary finding was that using fine-tuned models resulted in additional improvements in performance metrics against the annotated clinical data. In particular, higher recall rates indicate that the model is now able to detect kidney stones in challenging situations (i.e., false negative detections), while improved precision rates indicate that the model tends to produce fewer false positive detections. Overall, these findings suggest that this proposed approach generalizes well to real world scenarios with clinical patient data.

4.4.2 Detection Results

The image results of the object detection model are provided to show how well the models localize kidney stones in ultrasound images.

Figure 4.2 Detection sample using YOLOv8m model of Phase-I

The detection output for Phase I shows that the models accurately detect the stone locations in a controlled environment. The predicted bounding boxes closely match the ground truth annotations. However, there were some limitations associated with detecting very small stones and low-contrast regions which resulted in some false negatives or lower confidence scores.

The Phase II results show how robust the models are when used on real clinical ultrasound images. The models had great success localizing kidney stones despite encountering noise, variability and imaging artifacts. Furthermore, the improvement in

Phase II detection results as compared to those observed in Phase I supports the efficacy of model fine-tuning using clinical data.

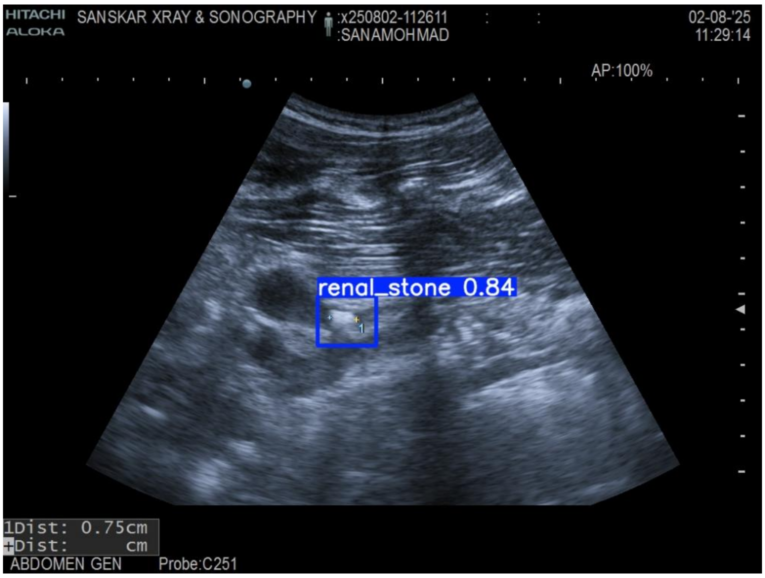


Figure 4.3 Detection sample using YOLOv8m model of Phase-II

4.4.3 Grad-CAM Results

The use of Grad-CAM allowed us to determine which parts of the image were impacting how the model predicted.

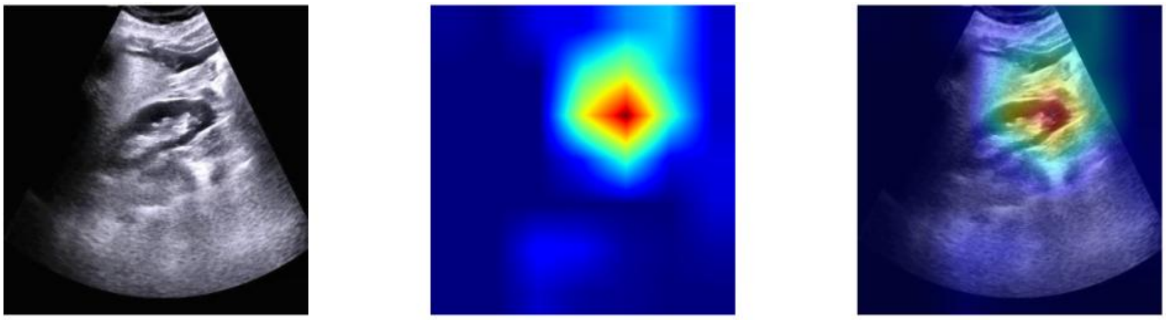


Figure 4.4 Grad-CAM result sample of Phase-I

The output from Grad-CAM gives us evidence that our model is looking at regions where kidney stones exist. Our heatmaps specifically give us region of interest, which provides us with confirmation that the model is identifying and learning important features.

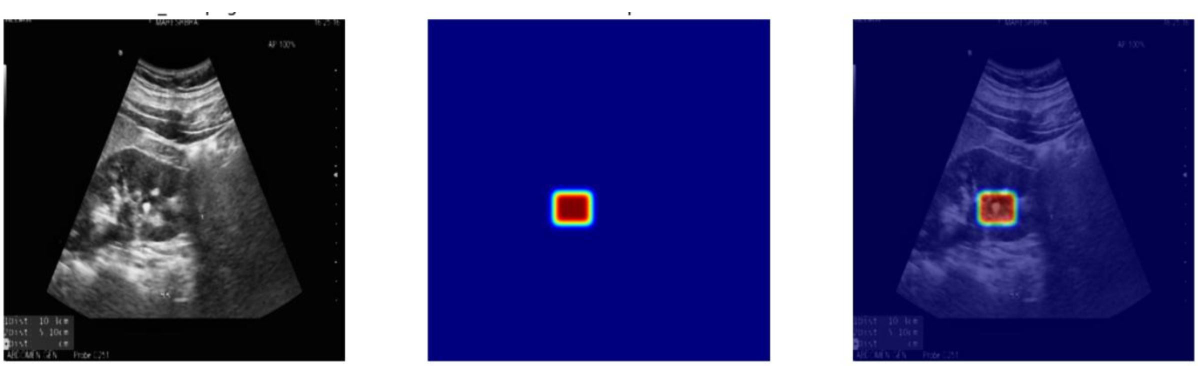


Figure 4.5 Grad-CAM result sample of Phase-II

The output for Phase II of the Grad-CAM gives us a more directed area of the kidney stones detected by the model. This indicates that the model has learned to operate in the field more accurately and makes more accurate predictions than in Phase I. The combination of detection and explanation will enhance the ability of clinicians to understand and interpret the results from the system.

4.4.4 Web page system outputs

To test the proposed renal stone detection system's effectiveness, a website was developed using an integrated platform that contains detection, visualization, and reporting features.

The user-visible interface is demonstrated in Figure 4.6. This interface shows all parts of the system (objects of use) and what it does. The user has an opportunity to comprehend how to operate it prior to actually utilizing the system.

As seen in Figure 4.7, the interaction process starts with a user uploading an ultrasound image. Multiple ultrasound images can be uploaded and once the upload is complete; the ultrasound image will follow a simple workflow consisting of upload, execution and review of the results. This structured design provides the user with an easy to use and clear process to follow when using this software.

After completing the upload and processing of an image, detection results are generated in the form of Figure 4.8 (as seen at the bottom of this page). The detected kidneys stones can be identified by either their bounding boxes showing where the stones were detected as seen on the original image or by being integrated into the Grad-CAM heat map.

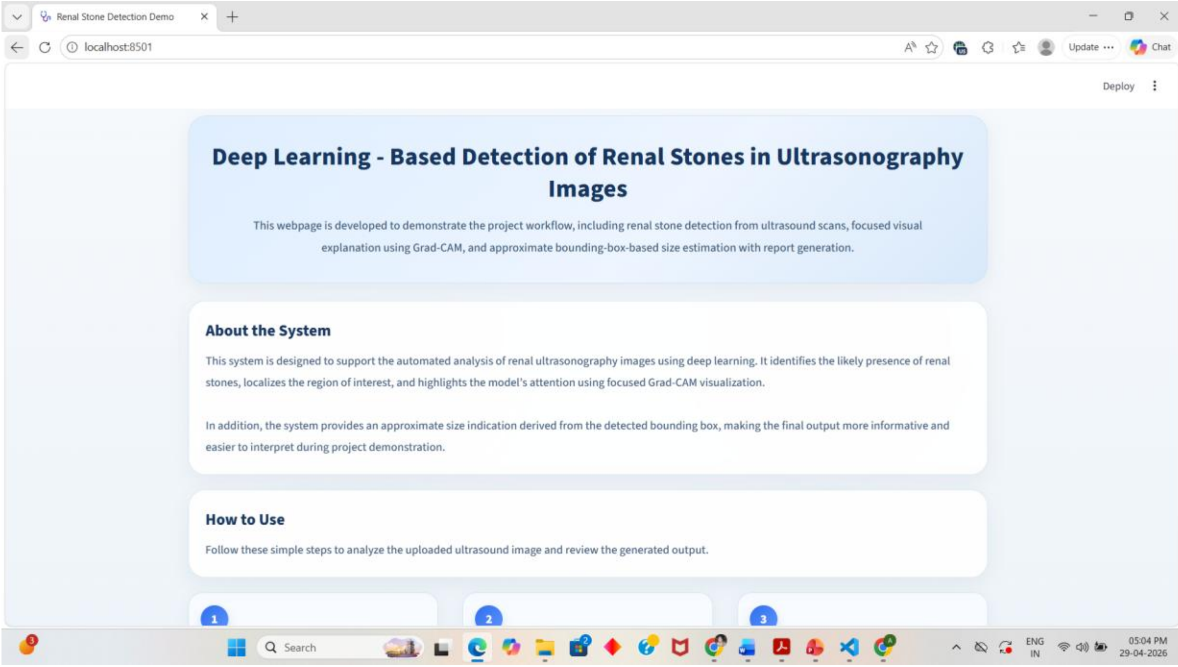


Figure 4.6 Web Interface Home page

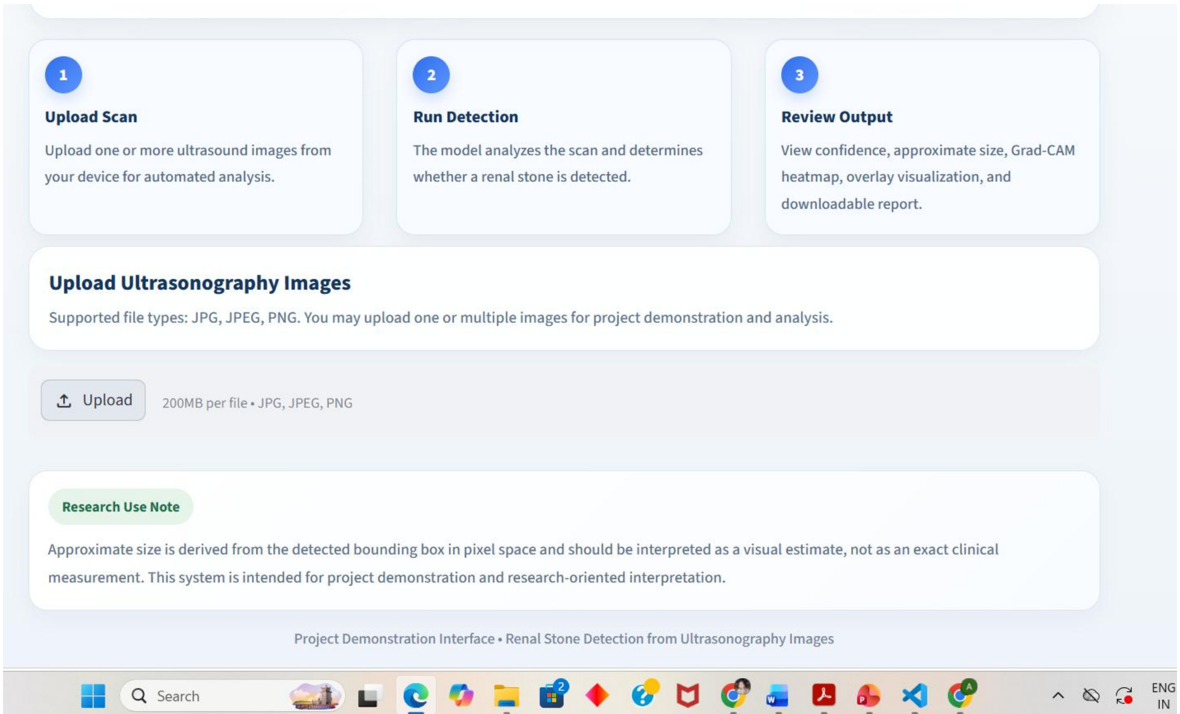


Figure 4.7 Image Upload Section

A summary of the prediction is also provided that contains a status of whether the kidney stones were detected with estimates of the volume of the identified kidney stones based on their bounding box sizes.

After a detection has been made, a structured report will be created automatically and sent to you. An example of this report is demonstrated in Figure 4.9 above. The report provides case data and relevant exam details as well as a summary of IDS findings that were generated from an artificial intelligence scan.

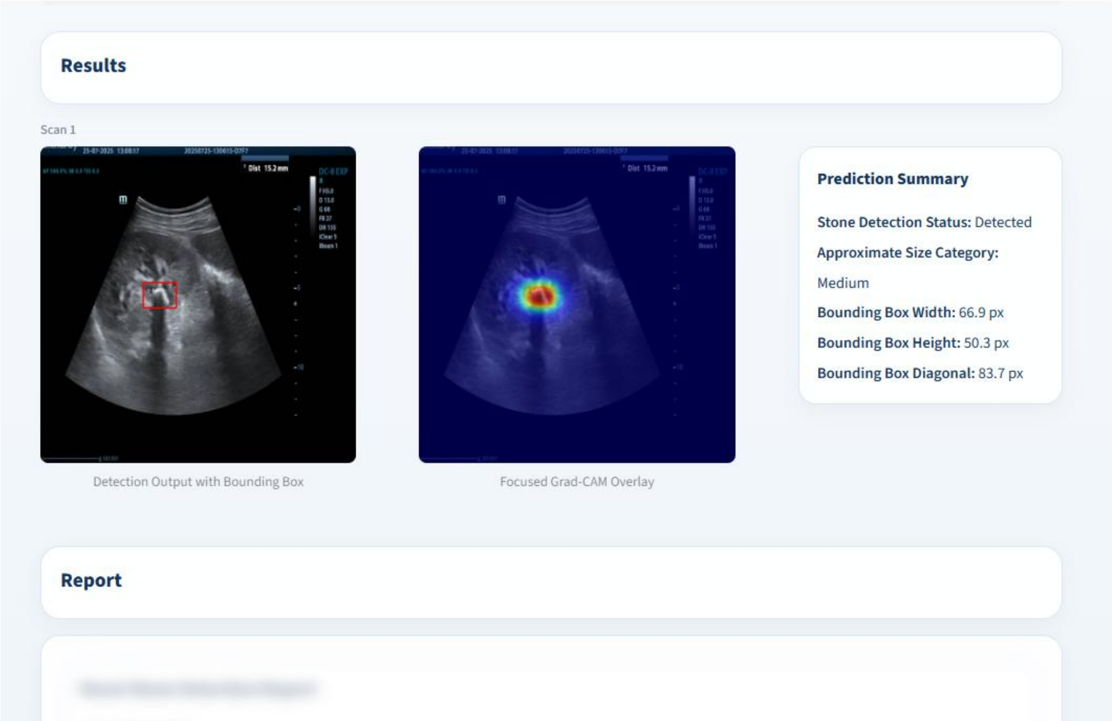


Figure 4.8 Detection output with Grad-CAM Visualization

Figure 4.10 shows both the detection and Grad-CAM outputs for this experiment; therefore, the user can compare their region of detection with the location of the model's attention. This should help the user to better understand the prediction output from the model.

The system shown in Figure 4.11 illustrates another example of how this model behaves systemically. When a renal stone does not exist, the model does accurately predict that there will not be renal stones (meaning that it does not generate any false bounding box), thus demonstrating it can avoid generating false positive predictions as well.

This result demonstrates that the algorithm is able to identify negative samples accurately while not producing any false detects. This fact is important in a clinical environment where the chances of false positive and subsequent action will be minimized. These data support the conclusion that this system can be counted on to perform consistently for both positive and negative cases.

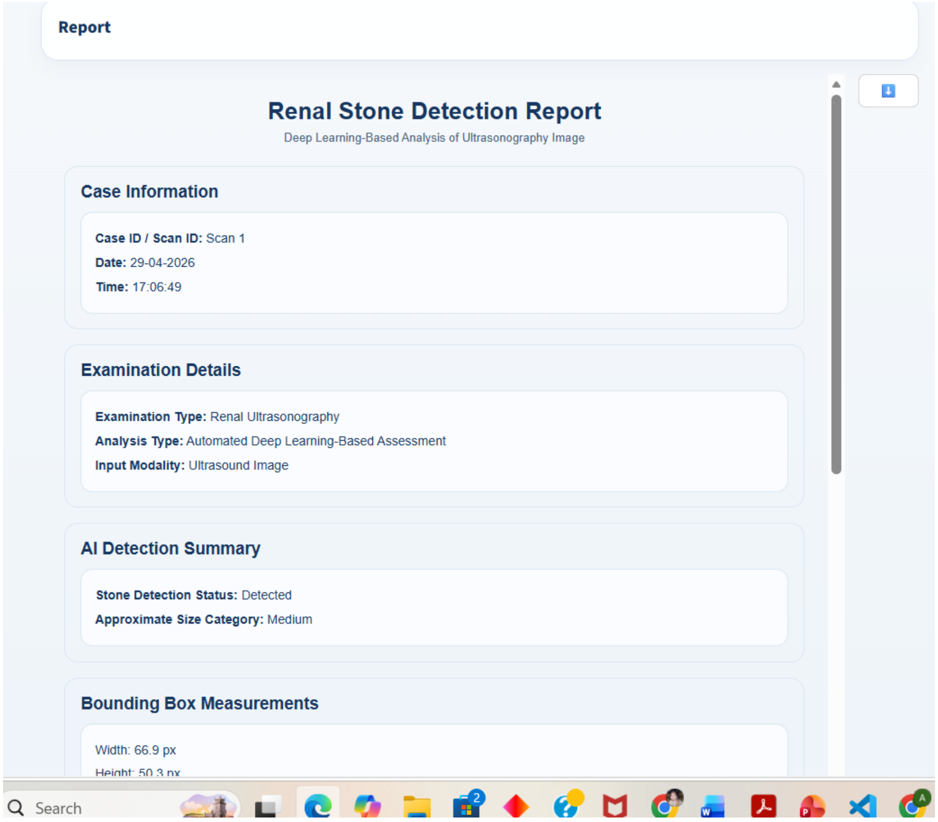


Figure 4.9 Generated Report

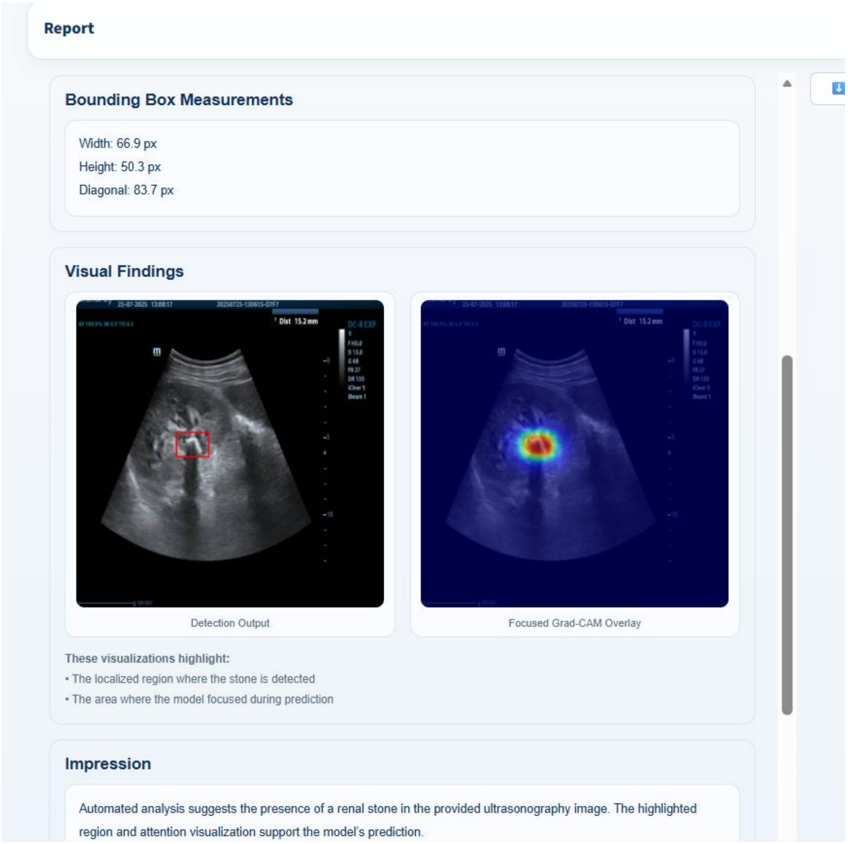


Figure 4.10 Generated Report with Detection Output Images

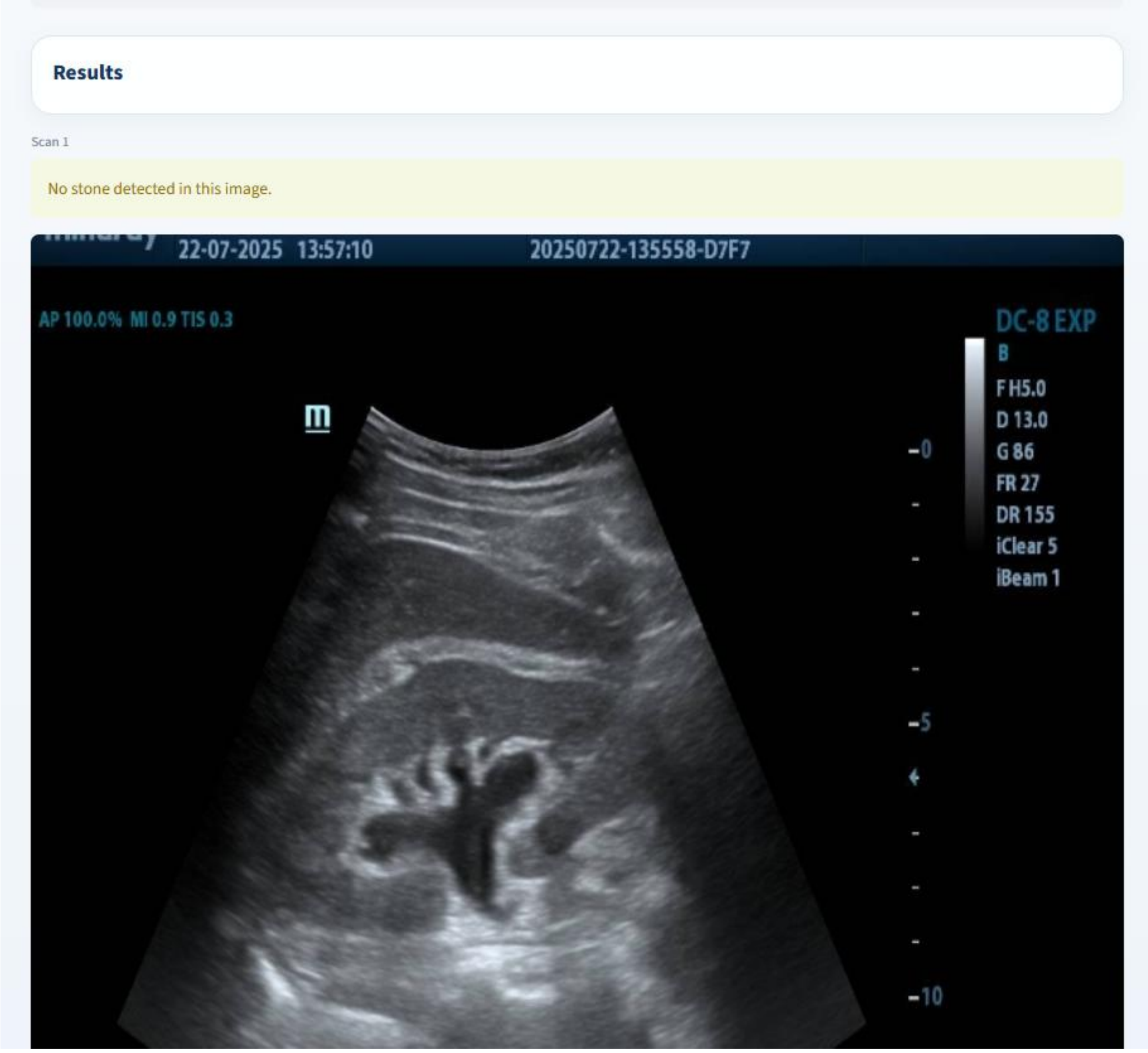


Figure 4.11 No Stone Detection Sample

These three capabilities combined in one interface, detection, explainability and reporting, represent a complete end to end solution. These capabilities show that the system can be deployed in actual clinical settings and serve as an effective decision-support system for medical imaging applications.

4.5 Comparative Results

Table 4.3 compiles a summary of the comparative performance between Phases I and II of the proposed renal stone detection system and illustrates an increase in model performance from using a controlled public dataset to using data collected during clinical practice in the real world.

The findings reveal a considerable enhancement in the performance of detection of kidney stones following fine-tuning of the models with clinical datasets. Precision increased from 0.822 (Phase 1) to 0.925 (Phase 2) indicating a decrease in false-positive detections. Similarly, recall was improved from 0.726 to 0.955 demonstrating the improved capacity of the model to accurately identify kidney stones under difficult imaging conditions.

Table 4.3: Performance comparison between Phase-I and Phase-II

Metric/ Aspect	Phase-I (Public Dataset)	Phase-II (Clinical Dataset)
Dataset Type	Public	Real clinical
Precision	0.822	0.925
Recall	0.726	0.955
Detection Performance (mAP50)	82.12%	97.98%
Inference Time	14.1ms	12.4ms
Robustness to noise	Moderate	High

Detection performance as measured by mAP@0.5 also exhibited a significant increase from 82.12% to 97.98%. This improvement demonstrates how well the model can localize kidney stones in clinical ultrasound images, despite the inherent challenges of ultrasound images (noise, low contrast, and variances in acquisition conditions).

An important finding was the decrease in inference time of the model as measured in Phase I and II of this study. The model developed during Phase I took 14.1 ms (the time for a single inference to be completed) whereas model developed in Phase II took 12.4 ms. Not only does this suggest improved accuracy, but also implies greater computational efficiency of the model, enabling the potential usage of this optimized model for real-time tasks.

Furthermore, the robustness of the model to noise and variability has increased considerably. The Phase I models were observed to exhibit moderate robustness, while Phase II models exhibited strong robustness, due to their manufacturing from realistic clinical data used during training and finetuning. This finding strongly supports the need for development and training using realistic dataset sources to produce reliable models to be utilized in the real world.

This comparative analysis shows that the proposed approach solves some of the problems found in literature concerning localization accuracy, robustness to noise, and practicality. Additionally, moving from Phase 1 to Phase 2 demonstrates that the proposed system is effective and has high potential for clinical implementation.

Chapter-5 Discussion

This chapter includes discussion of the results produced by the proposed system. An interpretation of the results, major findings, and analysis of the advantages and disadvantages.

5.1 Interpretation of Results

These experimental evaluation findings give valuable insight into how effectively and practically applicable the proposed renal stone detection system is. In addition to providing an indication of performance improvement, the results also quantify how the system performs in adapting to real-world medical imaging difficulties.

Perhaps the biggest change seen between Phases I and II was the increase in performance due to the use of clinical ultrasound data to fine-tuning the networks at the end of training in Phase II. Phase I allowed the networks to learn general features from a controlled dataset, while Phase II allows the networks to experience real-world variations like noise, low-contrast and inconsistent imaging conditions. This demonstrates that clinical data is essential to improving generalization and reliability of networks in real-world situations.

Based on its performance, the YOLOv8 detection model appears capable enough to meet the demands of medical imaging research requiring both accuracy and speed. In contrast to classification models that merely predict whether or not a given stone exists, detection models also give information about where on the image that stone is located. Because of this factor, they are much more useful to clinicians in determining what course of treatment to take with each patient than currently available classification-based technologies. This finding demonstrates the value of migrating from classification-based systems to detection-based systems when providing tools for practical medical use. Furthermore, the short amount of time needed to process an image with this model indicates that it could be used to provide almost real-time analysis as part of a clinical workflow.

The use of Grad-CAM in the system adds to its ability to develop visual explanations for how the model arrives at its predictions. The images created by Grad-CAM show that the model is concentrating on areas of the ultrasound scan that are pertinent to making predictions, thereby bolstering the trustworthiness of the model and mitigating a significant limitation of Deep Learning models, their lack of transparency. The level of precision achieved during phase II indicates that, following exposure to clinical data, the model has gained a greater degree of precision from the clinical data being processed.

One major feature of the results is how the system functions in cases where there is no renal stone. This shows that the model avoids false positives and therefore minimizes the anxiety associated with false positives. The model is sensitive to both types of cases, thus permitting a reasonable balance between sensitivity and specificity that is necessary for medical decision-support systems.

The findings imply that the system is functional as a reliable and trustworthy tool that can be analysed at both a logical level to demonstrate the functionality of its methods and data access attributes but also at an operational level through verification of results by other users to confirm they are within their normal operation parameters. The proven ability of the detection aspects of this system, when combined with the data available to support that claim, indicates that this system would be well received for use in a health care setting.

5.2 Key Findings

The findings from the evaluation of the experimental renal stone detection system were very important. For example, it is obvious from the experimental results that the use of object detection models has much greater utility than classification models. Although classification models are capable of identifying if renal stones are present in an image, the lack the ability to determine where in the image the renal stones actually exist. Object detection models can identify both the presence and the location of renal stones, thus making them better for use in clinical settings.

The data suggests that using clinical ultrasound data has resulted in an increase in model accuracy. The transition from phase I to phase II shows that exposure to real-world data improves a model's ability to generalize and perform well under difficult imaging conditions. This emphasizes the need for the use of clinically relevant data sets when developing medical imaging products.

An analysis comparing various object detection techniques indicated that YOLOv8-m offers the optimal combination of detection performance and speed. Although other more elaborate models provide similar overall levels of performance as those from YOLOv8-m; they do so at significantly greater times per image than YOLOv8-m's real-time performance. As a result of this finding, YOLOv8-m is considered ideal for both real-time and near-real-time situations.

The implementation of Grad-CAM has shown that it significantly and efficiently assisted with the model's interpretability because the generated heatmaps always indicated the areas of interest related to kidney stones, thus demonstrating visually to the user the decision-making process of the model. This builds user trust and makes it more acceptable for clinical use.

The other major finding was that the model efficiently identified situations where there were no renal stones, as no false results were placed in these cases. As no false results occurred in no stone cases, this indicates that the model is not over-sensitive, and it can adequately balance detection of true positives with the avoidance of false positive results. Both of these requirements are necessary for reliable decision-support systems in healthcare.

Finally, the ability to use a single web-based platform for detection, visualization, and reporting demonstrates that it can be practically implemented in the real world. The structured output generated by the web application and visually presenting the results in an easy-to-understand format shows that this is an end-to-end solution for detecting renal stones from ultrasound imagery.

5.3 Advantages and Limitations

There are many benefits to this renal stone-detection system; therefore, it could be very useful in medical imaging. One of the main benefits of this system is its capability of performing both identifying and locating kidney stones in ultrasound images. In contrast, a traditional classification approach does not give any spatial information; thus, a bounding box gives the spatial information required to diagnose and plan treatment clinically.

The use of authentic ultrasound data in Phase II is also a crucial advantage as this improves the reliability and generalization of the model. The training and fine-tuning of the model using real-situated data enable the system to become increasingly proficient at adjusting to the many factors that affect normal and pathological ultrasound imaging (i.e., noise, low contrast and variability).

This combination adds another layer of interpretability via Grad-CAM's explanatory function, providing visual feedback regarding the model's decision-making process. Transparency and trust are very important for many types of applications, but particularly in cases where the user has no prior exposure to the underlying technology (e.g., in a medical application).

Also, developing a web-based interface showed that the proposed system is possible to create. The system allows users to upload images, see the results of the detection and create a structured report all from one website. This shows how the proposed system can be used as an end-to-end solution and that it is easy to demonstrate and could be deployed.

There are some disadvantages of using this system, though it has benefits as well. One of the biggest drawbacks to this system is that all stones detected by this system are measured only in pixels; there are no real-world, unit measurements (like mm). Therefore, any estimated size measurements provided by this system must be viewed as an approximate visual representation instead of an exact clinical measurement.

A further limitation of this work is the small size of the clinical dataset. An improvement in both the applicability and robustness of the model may be realised with access to a substantially larger and more heterogeneous dataset. Furthermore, there is considerable variability in ultrasound imaging due to differences in equipment settings, operator techniques and patient conditions; all of which may impact the performance of the model in situations that were not part of the training data.

While the system shows excellent performance under controlled laboratory testing conditions, it has not been tested in the actual clinical workflow with actual medical personnel yet. Before it can be used for standard clinical practice, additional testing in actual clinical settings will need to happen.

Chapter-6 Conclusion and Future Work

6.1 Conclusion

This research has developed a complete method for using deep learning to automatically identify kidney stones using ultrasound images. This proposed approach incorporates classifier, object detection and explainable AI methods to assist with key challenges of ultrasound imaging, including background noise, reduced contrast, and variation in the way data is collected. The arrangement of these methods in a structured pipeline allows for both accurate identification and actionable insight from the predictions generated by the deep learning algorithm.

Experimental results show that object detection models (especially YOLOv8) can successfully identify and locate kidney stones in ultrasound images. The transition between Phase I and Phase II showed how important it is to include real world clinical data in the development of models, as this led to a significant improvement in performance and generalization. This demonstrates that reliable medical imaging systems require training with realistic datasets.

Furthermore, with additions to the model such as Grad-CAM, visual representations are able to illustrate how the model arrives at its predictions. Having interpretability with your deployed models increases user trust in predictions, and provides a solution to one of the main shortcomings of deep-learning based models when used for healthcare purposes. The ability of the system to accurately identify both positive and negative examples indicates that the system can be relied upon and used as a decision support mechanism.

An additional key finding of this study is the creation of an online platform that combines detection, viewing, and report generation all within one system. This complete implementation supports the technical viability of the proposed system while also demonstrating that it can be used in actual clinical settings where people go for treatment. A successful response has been given to most limitations identified by the current literature via the presented method. This includes the lack of localizing capabilities associated with classification algorithms; the low robustness of most classification algorithm's results (when using real world data), and there is currently no interpretability in most of the tables reported. All of the evaluated systems provided an acceptable combination of accuracy, efficiency and usability as a viable option for ultrasound renal stone image detection.

To conclude, the renal stone detection system mentioned above shows that the use of deep learning, explainability, and practical deployment can address real problems in medical imaging. The system can detect renal stones accurately and reliably as well as provide an explanation for how the detection was made, giving the healthcare professional some insight into what they're seeing. In this regard, we believe that through our work on this system, we are able to show examples of how AI technology can be used to enhance the diagnostic efficiency and improve the decision-making capability of professionals involved in the healthcare profession and help pave the way for better, more effective solutions that can be integrated into clinical environments in the future.

6.2 Future Scope and Enhancement

The renal stone detection system has been effective at detecting and locating renal stones. However, there are numerous possibilities for the system's continued improvement and augmentation. One primary opportunity will be to improve the estimation of the size of detected stones. Currently, the current system determines the size of detected stones using pixel measurements of the stone's bounding box, with no ability to convert those pixel measurements into real-world units (e.g., millimetres). Future work could thus focus on introducing calibration techniques to convert those bounding box measurements into clinically significant real-world measurements.

One way to improve the model's performance is by expanding the dataset size. The more large, diverse datasets are for ultrasound, which includes acquiring ultrasound images from different machine types, with many different types of patients, the more likely that the model will be able to generalize to any other real-world cases and will have a greater degree of robustness and ability to provide consistent output across multiple real-world cases.

There are many other ways to improve the architecture of models such as model architecture as well. By having one single model that does multiple jobs, like detecting and segmenting an object, the model may perform better since it can look at the underlying fine details in a spatial way. Another important consideration is to make sure that the model is optimized for performance so that it has a lower computational burden so that it can be deployed more quickly and with less cost/resources.

The system's explainability features have the potential to be improved upon as well. Though Grad-CAM provides a meaningful and visual way to understand model predictions, additional research could focus on exploring other alternative explainability techniques that produce a deeper understanding of the models. An increase in explainability will make the system more interpretable and build further confidence in the overall system.

Lastly, validate the system through partnerships with physicians in true clinical environments to confirm by performing clinical validation trials and providing expert input on the utility of the system and its further improvements. Transitioning from a research prototype to an acceptable clinician-based solution will require this last step.

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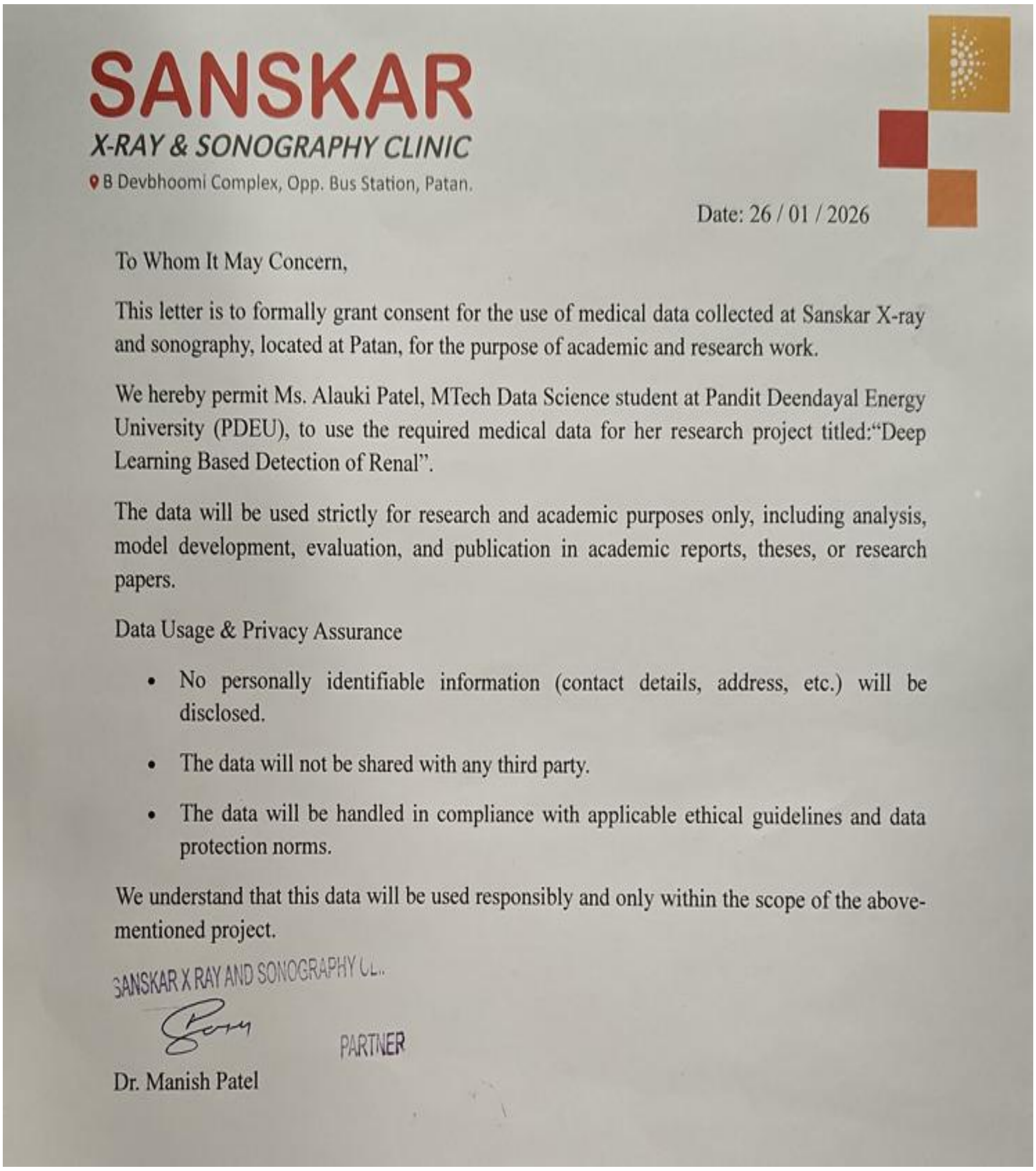
Appendix A: Consent Letter for Data Usage

Permission was given by Dr. Manish Patel (Radiologist, Sanskar X-Ray and Sonography Clinic, Patan) for the use of clinical ultrasound data in this study.



SANSKAR

X-RAY & SONOGRAPHY CLINIC

B Devbhoomi Complex, Opp. Bus Station, Patan.

Date: 26 / 01 / 2026

To Whom It May Concern,

This letter is to formally grant consent for the use of medical data collected at Sanskar X-ray and sonography, located at Patan, for the purpose of academic and research work.

We hereby permit Ms. Alauki Patel, MTech Data Science student at Pandit Deendayal Energy University (PDEU), to use the required medical data for her research project titled:"Deep Learning Based Detection of Renal".

The data will be used strictly for research and academic purposes only, including analysis, model development, evaluation, and publication in academic reports, theses, or research papers.

Data Usage & Privacy Assurance

- No personally identifiable information (contact details, address, etc.) will be disclosed.
- The data will not be shared with any third party.
- The data will be handled in compliance with applicable ethical guidelines and data protection norms.

We understand that this data will be used responsibly and only within the scope of the above-mentioned project.

SANSKAR X RAY AND SONOGRAPHY CL..

PARTNER

Dr. Manish Patel

Appendix B: Use of AI Tool

Figures 3.6 and 4.1 were generated with the assistance of ChatGPT for visualization purposes only. These figures are intended only for conceptual illustration and do not affect the experimental results or findings of the study.

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*% detected as AI

AI detection includes the possibility of false positives. Although some text in this submission is likely AI generated, scores below the 20% threshold are not surfaced because they have a higher likelihood of false positives.

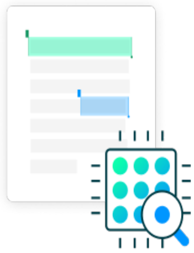
Caution: Review required.

It is essential to understand the limitations of AI detection before making decisions about a student's work. We encourage you to learn more about Turnitin's AI detection capabilities before using the tool.

Frequently Asked Questions

How should I interpret Turnitin's AI writing percentage and false positives?

The percentage shown in the AI writing report is the amount of qualifying text within the submission that Turnitin's AI writing detection model determines was either likely AI-generated text from a large-language model or likely AI-generated text that was likely revised using an AI paraphrase tool or word spinner.



False positives (incorrectly flagging human-written text as AI-generated) are a possibility in AI models.

AI detection scores under 20%, which we do not surface in new reports, have a higher likelihood of false positives. To reduce the likelihood of misinterpretation, no score or highlights are attributed and are indicated with an asterisk in the report (*%).

The AI writing percentage should not be the sole basis to determine whether misconduct has occurred. The reviewer/instructor should use the percentage as a means to start a formative conversation with their student and/or use it to examine the submitted assignment in accordance with their school's policies.

What does 'qualifying text' mean?

Our model only processes qualifying text in the form of long-form writing. Long-form writing means individual sentences contained in paragraphs that make up a longer piece of written work, such as an essay, a dissertation, or an article, etc. Qualifying text that has been determined to be likely AI-generated will be highlighted in cyan in the submission, and likely AI-generated and then likely AI-paraphrased will be highlighted purple.

Non-qualifying text, such as bullet points, annotated bibliographies, etc., will not be processed and can create disparity between the submission highlights and the percentage shown.